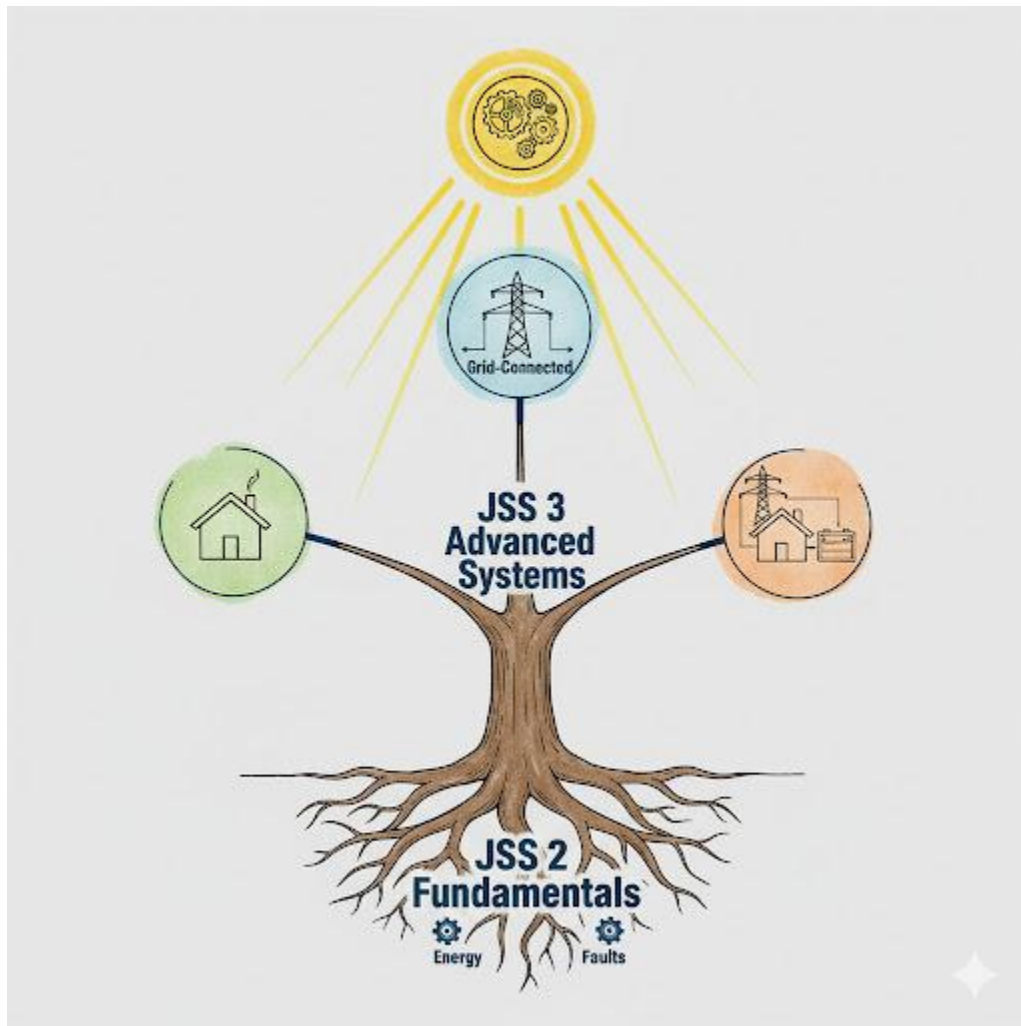


JSS 3 SOLAR PHOTOVOLTAIC (PV) INSTALLATION AND MAINTENANCE

LESSON NOTES

FIRST TERM

WEEK 1: INTRODUCTION AND REVISION OF PREVIOUS WORK



Content: (I) Recap of JSS 2 topics: energy storage, performance evaluation, and fault detection (II) Understanding advanced system configurations (III) Classroom discussion: evolution of solar technology.

Performance Objectives: By the end of this lesson, students should be able to:

1. Recall and discuss the functions of key solar components (panels, batteries, inverters, charge controllers).
2. Explain the concepts of system sizing, load estimation, and fault detection from JSS 2.
3. Differentiate between the three main system configurations (standalone, grid-tied, hybrid).
4. Discuss the evolution of solar technology from simple lighting to complex power systems.

Lesson Content:

1. Review of JSS 2 Concepts This first week is a crucial recap to ensure all students are ready for the advanced topics in JSS 3.

- **A. Key Components (Recap):**

- **Solar Panels (PV Modules):** Convert sunlight (photons) directly into DC electricity.
- **Batteries (Energy Storage):** Store the DC electricity for use at night or on cloudy days. (Reviewed *Energy Storage*).
- **Charge Controller:** The "gatekeeper" that protects the battery. It regulates the voltage and current from the panels to prevent *overcharging* and *deep discharging*.
- **Inverter:** The "translator." It converts the low-voltage DC electricity from the batteries into high-voltage AC electricity used by most household appliances (TVs, fans, fridges).

- **B. System Performance & Faults (Recap):**

- **Performance Evaluation:** Checking if the system is producing the power it is supposed to. This involves measuring **Voltage (V)** and **Current (A)** with a multimeter.
- **Fault Detection:** Identifying problems. Common JSS 2 faults included:
 - *Panel shading* (leading to low power output).
 - *Loose connections* (a major fire hazard).
 - *Battery failure* (due to age or poor maintenance).
 - *Inverter overload* (trying to power too many appliances).

2. Understanding Advanced System Configurations As a review, we will re-examine the three main ways a solar PV system can be set up.

- **A. Standalone (Off-Grid) System:**

- **Definition:** The most common type in Nigeria. The system is 100% independent of the national grid (e.g., PHCN/NEPA). It generates, stores, and uses its own power.
- **Components:** Solar Panels + Charge Controller + Battery Bank + Inverter (for AC loads).
- **Pros:** Complete energy independence. No utility bills. Works in remote areas.
- **Cons:** Requires an expensive battery bank. Limited by the amount of power you can store.

- **B. Grid-Tied (On-Grid) System:**

- **Definition:** A system that is connected to the national grid. It does **not** have batteries.
- **How it works:** During the day, it uses solar power. If it produces *extra* power, it can "sell" it back to the grid. At night, it *buys* power from the grid.
- **Pros:** Cheaper (no batteries). Highly efficient.
- **Cons:** Does *not* work during a power outage (it shuts down for safety). Requires reliable grid.

- **C. Hybrid System:**

- **Definition:** The "best of both worlds." It is a grid-tied system that *also* includes a battery bank.
- **How it works:** It can use solar power, store it in batteries, *and* draw power from the grid. It can be programmed to:
 1. Use solar power first.
 2. Use battery power second (e.g., at night).
 3. Use grid power last (only as a final backup).
- **Pros:** Provides reliable power 24/7 (energy security) and can also reduce electricity bills.
- **Cons:** The most complex and expensive configuration.

3. Discussion: The Evolution of Solar Technology

- **Early Days (1970s-1990s):** Solar was very expensive. It was mostly used for small, "niche" applications like space satellites or simple solar-powered calculators.
- **JSS 1 Level (Simple Systems):** Focused on small, DC-only systems (e.g., a small panel charging a phone or powering a single 12V DC bulb).
- **JSS 2 Level (Home Systems):** Introduced inverters and larger battery banks, allowing for the powering of AC loads like fans and TVs in a standalone setup.
- **JSS 3 Level (Advanced Systems):** We now see the "evolution" into *smarter* systems:
 - **Smarter Controllers:** The shift from basic **PWM** controllers to highly efficient **MPPT** (Maximum Power Point Tracking) controllers.
 - **Smarter Batteries:** The move from heavy, maintenance-heavy **Lead-Acid** batteries to lightweight, long-lasting, "smart" **Lithium-ion** batteries.
 - **Smarter Systems:** The rise of **Hybrid** and **Grid-Tied** systems that can "think" and interact with the grid, not just act as a simple generator.

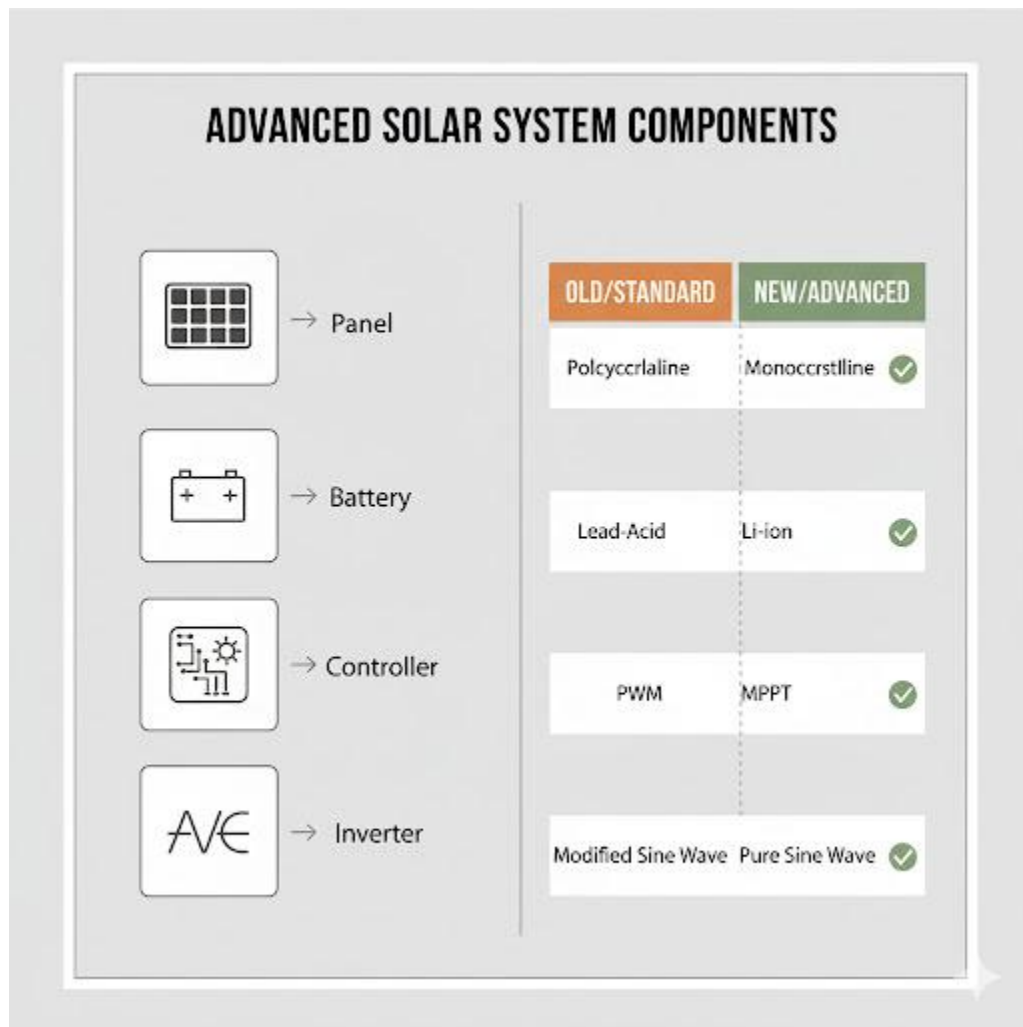
Evaluation:

1. What is the primary function of an inverter?
2. What is the main difference between a "standalone" and a "grid-tied" system?
3. Why is a "hybrid" system more reliable than a "grid-tied" system?
4. What was a common fault we learned to detect in JSS 2?
5. What is one way solar battery technology has "evolved"?

Assignment:

1. Draw a simple block diagram for all three (3) system configurations: (a) Standalone, (b) Grid-Tied, and (c) Hybrid.
2. In your own words, what is the job of a Charge Controller?
3. Why is it important to check a system for "panel shading"?
4. What do you think is the biggest advantage of a hybrid system for a Nigerian home?
5. What does "MPPT" stand for, and why is it better than PWM? (Requires brief research).

WEEK 2: ADVANCED SOLAR SYSTEM COMPONENTS



Content: (I) Detailed study of PV modules, batteries, inverters, and charge controllers (II) Latest innovations (Li-ion batteries, MPPT controllers) (III) Component identification and selection (IV) Practical comparison.

Performance Objectives: By the end of this lesson, students should be able to:

1. Differentiate between monocrystalline and polycrystalline PV modules.
2. Explain the significant advantages of Lithium-ion (Li-ion) batteries over Lead-Acid batteries.
3. Explain the functional difference and benefit of an MPPT controller vs. a PWM controller.

4. Explain the difference between a "pure sine wave" and "modified sine wave" inverter.
5. Select appropriate components for a specific installation scenario.

Lesson Content:

1. Detailed Study of PV Modules (Solar Panels)

- **Recap:** A PV module is made of many solar *cells* linked together.
- **Types of Crystalline Silicon Panels:**
 - **A. Monocrystalline Panels:**
 - **Appearance:** Black in color, with the cells having "cut-off" rounded corners.
 - **How it's made:** Made from a *single, pure* silicon crystal.
 - **Pros: Highest efficiency** (18-24%). Performs better in high heat and low light. Lasts the longest.
 - **Cons: Most expensive.**
 - **B. Polycrystalline Panels:**
 - **Appearance:** Blue, "marbled" or "flaky" look, with perfectly square cells.
 - **How it's made:** Made from *multiple* silicon crystals melted together.
 - **Pros: Cheaper** than monocrystalline. Good performance.
 - **Cons: Less efficient** (15-18%). Less tolerant of high temperatures.

2. Latest Innovations: Batteries (Lead-Acid vs. Lithium-ion)

- **A. Lead-Acid Batteries (Old Technology):**
 - **Types:** Flooded (requires adding distilled water) and Sealed (AGM/Gel).
 - **Pros:** Low cost (upfront), widely available, reliable if maintained.
 - **Cons:** Very heavy, requires ventilation (releases gas), **low Depth of Discharge (DoD) - you can only use ~50% of the energy**, shorter lifespan (3-5 years).
- **B. Lithium-ion (LiFePO₄) (New Technology):**
 - **Definition:** The same technology used in your smartphone and electric cars, now adapted for solar.
 - **Pros:**

1. **High DoD:** You can safely use 80-100% of the battery's energy.
 2. **Long Lifespan:** Lasts 10-15+ years (5,000+ cycles vs. 500 for lead-acid).
 3. **Lightweight:** 1/3 the weight of lead-acid.
 4. **No Maintenance:** No gassing, no adding water.
 5. **Fast Charging:** Can charge to full in 2-3 hours.
- **Cons:** Very high upfront cost (but cheaper over its lifetime).

3. Latest Innovations: Charge Controllers (PWM vs. MPPT)

- **A. PWM (Pulse Width Modulation) (Old Technology):**
 - **Function:** A simple, switch-based controller. It just *cuts off* the panel's voltage to match the battery's voltage.
 - **Pros:** Very cheap, durable, good for small DC-only systems.
 - **Cons:** Very inefficient. It wastes a significant amount of power, especially if the panel voltage is much higher than the battery voltage.
- **B. MPPT (Maximum Power Point Tracking) (New Technology):**
 - **Function:** A "smart" controller. It is a sophisticated converter that *finds* the *perfect* voltage/current combination (the "maximum power point") from the panel. It then *converts* the high panel voltage (e.g., 60V) down to the battery voltage (e.g., 12V) *while increasing the current*.
 - **Benefit ($V_{\text{panel}} \times A_{\text{panel}} = V_{\text{battery}} \times A_{\text{battery}}$):** It "harvests" **15-30% more power** from the same solar panels.
 - **Pros:** Highest efficiency, essential for large or high-voltage systems.
 - **Cons:** More expensive.

4. Advanced Inverters (Pure Sine Wave vs. Modified Sine Wave)

- **A. Modified Sine Wave Inverter (Old/Cheap):**
 - **Output:** Produces a "blocky," stepped wave. It is a "choppy" approximation of AC power.
 - **Pros:** Very cheap.

- **Cons:** Can damage or destroy sensitive electronics (e.g., modern TVs, laptops, fridges, medical equipment). Makes fans and motors "buzz."
- **B. Pure Sine Wave (PSW) Inverter (Modern Standard):**
 - **Output:** Produces a smooth, clean wave that is *identical* or *better* than grid power.
 - **Pros: Safe for all appliances.** Runs motors and fans quietly and efficiently.
 - **Cons:** More expensive than modified.
 - **Rule:** For any serious installation (home or office), **always** use a Pure Sine Wave inverter.

5. Practical: Component Identification

- (In class, the teacher shows different components):
 - "This panel is blue and has square cells. Is it Mono or Poly?" (Poly)
 - "This battery is very heavy and has caps for water. What type is it?" (Flooded Lead-Acid)
 - "This inverter says 'PSW' on it. What does that mean?" (Pure Sine Wave)
 - "This controller is large, has a digital screen, and costs N50,000. Is it likely PWM or MPPT?" (MPPT)

Evaluation:

1. What is the visual difference between a monocrystalline and a polycrystalline panel?
2. What does "MPPT" stand for?
3. Why is a Lithium-ion battery better than a Lead-Acid battery? (Give 2 reasons).
4. Why should you use a "Pure Sine Wave" inverter instead of a "Modified" one?
5. Which controller (PWM or MPPT) is more efficient and "harvests" more power?

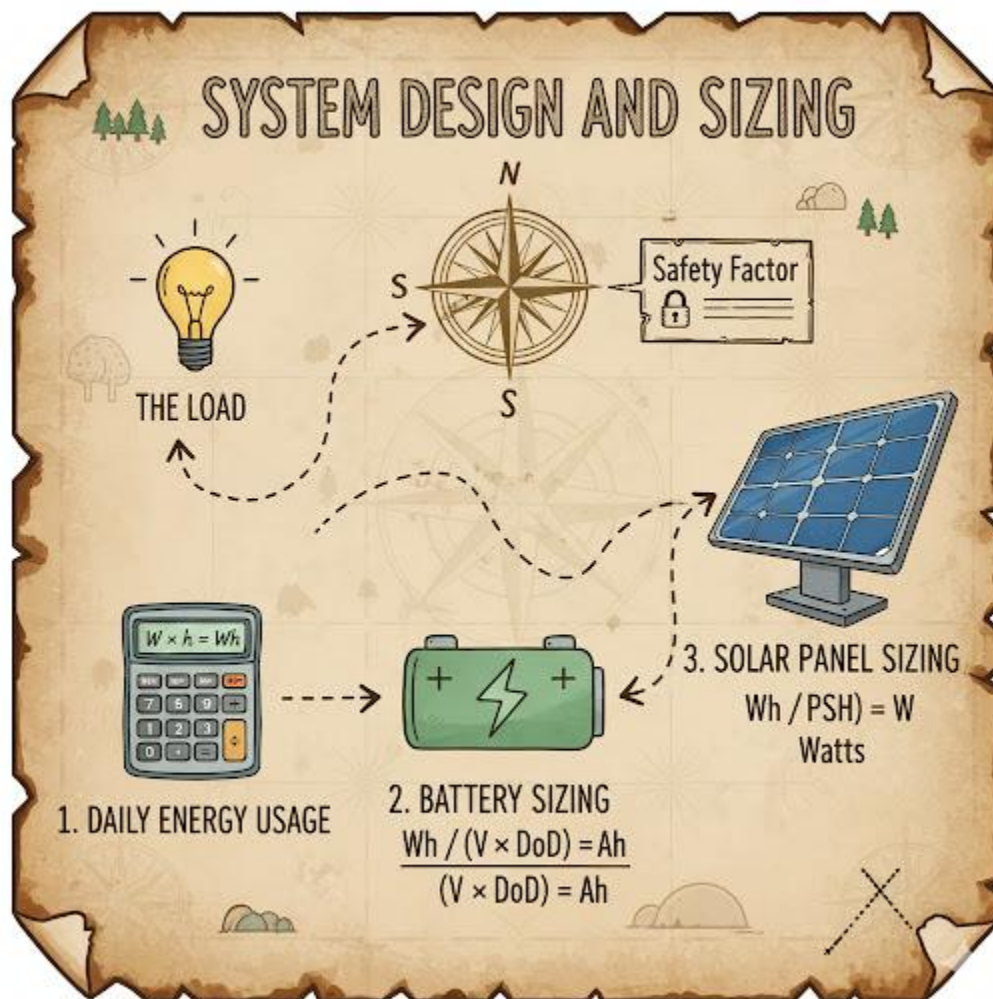
Assignment:

1. Define: Monocrystalline, Polycrystalline, MPPT, and Pure Sine Wave.
2. A customer wants to power their expensive laptop, new flat-screen TV, and fridge. Which *type* of inverter must you recommend?

3. A student needs a very small, cheap system for a single 12V bulb and to charge a phone.
Which *type* of charge controller would be most "cost-effective"?
4. What are two major "Cons" of using traditional Lead-Acid batteries?
5. Research: What does "DoD" stand for in relation to batteries, and why is a 90% DoD better than a 50% DoD?



WEEK 3: SYSTEM DESIGN AND SIZING



Content: (I) Principles of system design (II) Steps in sizing a solar system (III) Calculating load requirements (AC/DC loads) (IV) Sizing panels, batteries, and inverters (V) Hands-on activity: Designing a 12V DC lighting system.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Load Estimation" and "System Sizing."
2. Calculate the "Total Watt-hours" (Wh) of energy needed per day for a simple load.
3. Correctly size an inverter based on the "Total Watts" (W).
4. Correctly size a battery bank based on the "Total Watt-hours" (Wh) and "Days of Autonomy."

5. Correctly size the solar panel array needed to charge the battery bank.

Lesson Content:

1. Principles of System Design

- **Definition:** "System Sizing" is the most critical step in solar installation. It is the process of *calculating* the exact size of each component (panels, battery, inverter, controller) needed to meet a customer's specific energy needs.
- **The Golden Rule:** The entire system is built around the **Load**. You *must* know what the customer wants to power before you can do anything else.
- **The Goal:** To design a balanced system that is *reliable* (doesn't shut down), *durable* (doesn't spoil components), and *cost-effective* (not oversized).

2. Step 1: Calculating the Load Requirement (Load Estimation)

- **A. Identify all Loads:** List every single appliance the customer wants to power.
 - *Example:* 4 Bulbs, 1 Fan, 1 TV.
- **B. Find the Power Rating (Watts, W):** Check the sticker on the back of each appliance for its power consumption in **Watts (W)**.
 - *Example:* Bulb = 10W each, Fan = 60W, TV = 100W.
- **C. Estimate Daily Hours of Use (h):** Ask the customer how long they use each item per day.
 - *Example:* Bulbs = 5 hours/day, Fan = 8 hours/day, TV = 4 hours/day.
- **D. Calculate Energy per Day (Watt-hours, Wh):**
 - **Formula:** Energy (Wh) = Power (W) x Time (h)
 - *Load 1 (Bulbs):* $(4 \times 10W) \times 5h = 40W \times 5h = \mathbf{200\ Wh/day}$
 - *Load 2 (Fan):* $60W \times 8h = \mathbf{480\ Wh/day}$
 - *Load 3 (TV):* $100W \times 4h = \mathbf{400\ Wh/day}$
- **E. Calculate Total Daily Energy Need:** Add them all up.
 - *Total Wh/day:* $200 + 480 + 400 = \mathbf{1,080\ Wh/day}$

3. Step 2: Sizing the Inverter (The "Power" Component)

- The inverter must be able to handle the *total power* if all loads are turned on *at the same time*.
- **Formula:** Total Watts = (10W x 4) + 60W + 100W = 40W + 60W + 100W = 200W
- **Safety Factor:** Always add a 25-50% safety margin.
 - Inverter Size = Total Watts x 1.5
 - Inverter Size = 200W x 1.5 = 300W
- **Selection:** You must buy an inverter that is **at least 300W**. (e.g., a 300W, 500W, or 1000W model).
- **Voltage:** You must also choose a system voltage (e.g., 12V, 24V, 48V). For this small system, **12V** is fine.

4. Step 3: Sizing the Battery Bank (The "Energy" Component)

- The battery must be able to store the *total energy* (1,080 Wh) and handle no-sun days.
- **A. Account for Days of Autonomy:** How many days should the system run with *no sun*? (Standard = 1 or 2 days). Let's use 1 day.
- **B. Account for Battery Inefficiency & DoD:** Batteries are not 100% efficient. We will use a **DoD of 50% (for Lead-Acid)**.
 - **Formula:** Battery Bank (Wh) = (Total Wh/day x Days of Autonomy) / DoD
 - Battery Bank (Wh) = (1,080 Wh x 1) / 0.50 = 2,160 Wh
- **C. Convert Wh to Amp-hours (Ah):**
 - **Formula:** Amp-hours (Ah) = Watt-hours (Wh) / System Voltage (V)
 - Ah = 2,160 Wh / 12V = 180 Ah
- **Selection:** You must buy a **12V battery (or batteries) with a total of 180Ah**. (e.g., one 12V 200Ah battery).

5. Step 4: Sizing the Solar Panel Array (The "Charger")

- The panels must be able to fully recharge the *daily used energy* (1,080 Wh) in one day.
- **A. Find Peak Sun Hours (PSH):** How many hours of "good" sunlight does your location get? In Nigeria, this is about **4.5 to 5.5 hours**. Let's use **5 hours**.
- **B. Account for System Losses:** (Dust, heat, wires, etc.) about 30%.

- Panel Size (W) = Total Wh/day / (PSH x Efficiency_Factor)
- Panel Size (W) = 1,080 Wh / (5h x 0.70) = 1,080 / 3.5 = 308.5 W
- **Selection:** You must buy panels that total **at least 309W**.
 - *Option 1:* Two (2) x 160W panels (Total 320W).
 - *Option 2:* Four (4) x 80W panels (Total 320W).

6. Hands-On Activity: Designing a 12V DC Lighting System

- **Load:** 3 DC Bulbs (12V, 7W each)
- **Hours:** 6 hours/day
- *Step 1 (Load):* (3 x 7W) x 6h = 21W x 6h = **126 Wh/day**
- *Step 2 (Inverter):* **Not needed!** It's a DC-only system.
- *Step 3 (Battery):* (126 Wh x 1 day) / 0.5 DoD = 252 Wh.
 - Ah = 252 Wh / 12V = 21 Ah.
 - *Selection:* One small 12V 30Ah battery.
- *Step 4 (Panel):* 126 Wh / (5h x 0.70) = 126 / 3.5 = **36W**.
 - *Selection:* One 40W or 50W solar panel.
- *Step 5 (Controller):* Current (A) = Power (W) / Voltage (V) -> 40W / 12V = 3.3A. A small 5A or 10A PWM controller is perfect.

Evaluation:

1. What is "System Sizing"?
2. What is the *first* and most important step in sizing any solar system?
3. What is the difference between "Watts (W)" and "Watt-hours (Wh)"?
4. Why do we add a "safety factor" when sizing an inverter?
5. What does "Days of Autonomy" mean?

Assignment: You are asked to design a small system for a student.

- **Load:**
 - 1 Laptop (65W) used for 3 hours/day
 - 2 Bulbs (10W each) used for 4 hours/day

- **Calculate:**

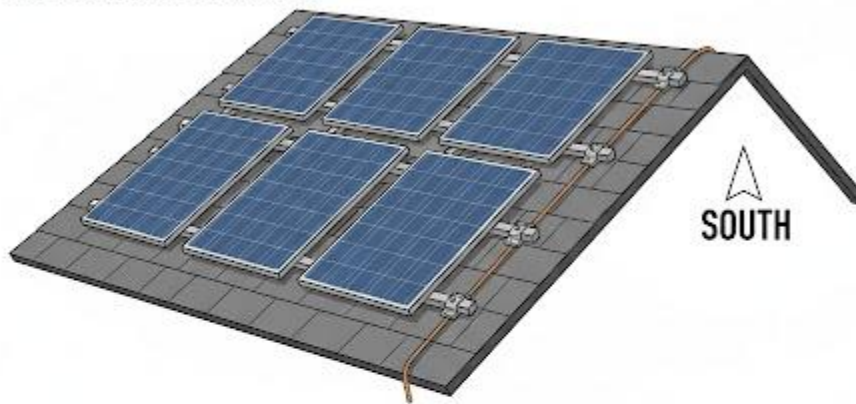
1. The laptop's daily energy use (in Wh).
2. The bulbs' total daily energy use (in Wh).
3. The **Total Daily Energy (Wh)** for the whole system.
4. The minimum **Inverter Size (W)** needed (use a 1.5 safety factor).
5. The **Battery Bank Size (Ah)** needed for a 12V system (assume 1 day autonomy and 50% DoD).



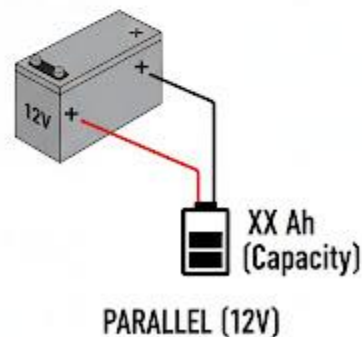
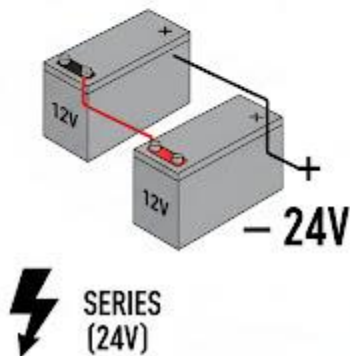
WEEK 4: SOLAR INSTALLATION PROCEDURES (ADVANCED)

SOLAR INSTALLATION PROCEDURES (ADVANCED)

ROOF PANEL INSTALLATION



BATTERY WIRING CONFIGURATIONS



Content: (I) Review of tools and materials (II) Mounting techniques (III) Series and parallel connections (practical) (IV) Grounding and cable management (V) Demonstration: Complete setup.

Performance Objectives: By the end of this lesson, students should be able to:

1. List the essential tools and materials for a solar installation.
2. Describe the process of securely mounting a solar panel on a roof.
3. Demonstrate a "series connection" with batteries or panels and state its effect (voltage adds up).

4. Demonstrate a "parallel connection" with batteries or panels and state its effect (current adds up).
5. Explain the critical importance of "grounding" (earthing).

Lesson Content:

1. Review of Tools and Materials

- **Safety Tools:** PPE (gloves, safety glasses, covered shoes).
- **Hand Tools:** Pliers, wire stripper, crimping tool, screwdrivers, spanners/ratchet set, hammer.
- **Measurement Tools:** Multimeter, tape measure.
- **Materials:** Solar panels, batteries, inverter, charge controller, PV cables (UV-resistant), battery cables, lugs, cable trunking (conduits), MC4 connectors, mounting rails and clamps.

2. Mounting Techniques for Panels Panels must be mounted securely to resist wind/rain and at the correct angle to catch the sun.

- **Key Principles:**
 - **Direction (Azimuth):** In Nigeria (Northern Hemisphere), panels must *always* face **South** to catch the maximum sunlight.
 - **Angle (Tilt):** The angle depends on your latitude. A good rule of thumb for Nigeria is a fixed tilt angle of **10-15 degrees**.
- **Mounting Steps (Roof Mount):**
 1. **Inspect Roof:** Ensure the roof is strong enough.
 2. **Locate Rafters:** Find the strong wooden beams under the roofing sheet.
 3. **Install Standoffs/Feet:** Drill and bolt the mounting "feet" *through* the roof and *into* the rafters, using sealant (like silicon) to prevent leaks.
 4. **Attach Rails:** Attach the long aluminum "rails" to the feet.
 5. **Mount Panels:** Place the solar panels onto the rails.

6. **Secure Clamps:** Use "mid-clamps" (between panels) and "end-clamps" (at the ends) to bolt the panel frames securely to the rails.
7. **Ground Panels:** Connect a grounding (earthing) wire to the frame of *each* panel.

3. Practical: Series vs. Parallel Connections This is the most important concept for wiring panels and batteries.

- **System Voltage:** 12V
- **A. Series Connection ("Chaining")**
 - **How:** Connect **Positive (+) to Negative (-)**.
 - **Effect:** **VOLTAGE adds up**, Current (Amps) stays the same.
 - *Panel Example:* Two 12V / 5A panels in **series** = **24V** / 5A system.
 - *Battery Example:* Two 12V / 100Ah batteries in **series** = **24V** / 100Ah battery bank.
 - **Use:** Used to *increase* the system voltage to match a 24V or 48V inverter.
- **B. Parallel Connection ("Jumping")**
 - **How:** Connect **Positive (+) to Positive (+)** and **Negative (-) to Negative (-)**.
 - **Effect:** **CURRENT (Amps) adds up**, Voltage stays the same.
 - *Panel Example:* Two 12V / 5A panels in **parallel** = 12V / **10A** system.
 - *Battery Example:* Two 12V / 100Ah batteries in **parallel** = 12V / **200Ah** battery bank.
 - **Use:** Used to *increase* the capacity (energy storage or charging power) while keeping the voltage the same.

4. Grounding and Cable Management

- **A. Grounding (Earthing):**
 - **Definition:** The most important safety feature. It is a dedicated wire that connects all the metal *frames* of your equipment (panels, inverter, mounts) to a long copper "earthing rod" hammered deep into the ground.
 - **Purpose:** If a wire insulation fails and the metal frame becomes "live" (electrified), the grounding wire provides a *safe path* for the dangerous electricity

to go directly into the earth. This will trip a breaker and *prevent you from being electrocuted*.

- **B. Cable Management:**

- **Definition:** Using tools to organize and protect all the wires.
- **Tools:**
 - **Trunking (Conduit):** Plastic pipes that protect wires from being cut or damaged by sun/rain.
 - **Cable Ties:** To bundle wires neatly.
- **Importance:** A neat installation is a safe installation. It prevents loose wires, short circuits, and makes future maintenance easy.

5. Demonstration: Complete Setup

- (Teacher and students connect a simple system in the lab):
 1. **Panel -> Charge Controller** (Connect the PV input terminals)
 2. **Controller -> Battery** (Connect the Battery output terminals. *This must be done BEFORE the panel*).
 3. **Battery -> Inverter** (Connect the heavy-duty battery cables to the inverter).
 4. **Inverter -> AC Load** (Plug a fan into the inverter).
 5. **Test:** Turn on the inverter and see the fan spin. Use a multimeter to measure panel voltage and battery voltage.

Evaluation:

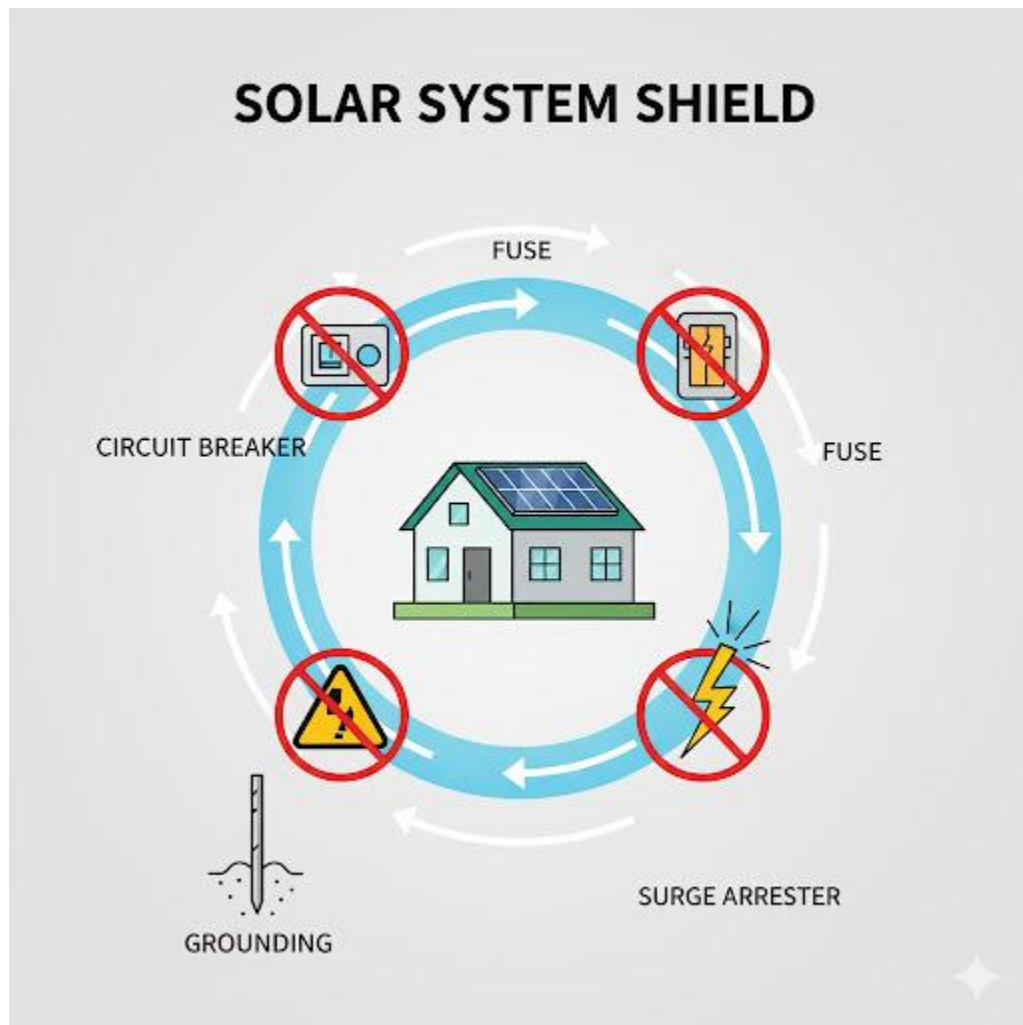
1. In Nigeria, which direction should solar panels face? (South).
2. What is the main purpose of "grounding" (earthing)?
3. You connect two 12V / 100Ah batteries in **series**. What is the new Voltage and Ah? (24V, 100Ah).
4. You connect the same two batteries in **parallel**. What is the new Voltage and Ah? (12V, 200Ah).
5. What is the difference between a "mid-clamp" and an "end-clamp"?

Assignment:

1. Draw a clear diagram of two 12V batteries connected in **series**.
2. Draw a clear diagram of two 12V batteries connected in **parallel**.
3. Why must you connect the charge controller to the *battery* **before** you connect it to the *solar panel*? (Research this).
4. What is "cable management" and why is it important?
5. A customer has a 24V inverter, but you only have 12V/200Ah batteries. How would you connect two batteries to power this inverter? (Draw the diagram).



WEEK 6: SYSTEM PROTECTION AND SAFETY DEVICES



Content: (I) Importance of safety (II) Protective devices: fuses, circuit breakers, surge arresters (III) Proper earthing/grounding (IV) Safe handling of tools and batteries (V) Practical: Installing a DC circuit breaker.

Performance Objectives: By the end of this lesson, students should be able to:

1. State the "Golden Rule" of electrical safety.
2. Differentiate between a "fuse" and a "circuit breaker" and state their purpose.
3. Define "surge arrester" and its function.
4. Re-emphasize the importance of "grounding" as a safety device.
5. Demonstrate how to safely connect a DC circuit breaker into a circuit.

Lesson Content:

1. Importance of Safety

- Solar PV systems are not toys. They can produce dangerous (even lethal) high-voltage DC electricity.
- **Golden Rule: "Safety First."** Never work on a "live" system. Always **disconnect** the power sources in the correct order (Breakers -> Panels -> Battery) before you begin work.
- **Main Hazards:**
 1. **Electric Shock:** From high-voltage DC (panels) or AC (inverter).
 2. **Fire:** From short circuits, loose connections ("arcing"), or overloaded wires.
 3. **Battery Hazards:** Lead-acid batteries contain corrosive acid and can release explosive hydrogen gas.

2. Protective Devices: Fuses and Circuit Breakers

- **Purpose (Overcurrent Protection):** The job of a fuse or breaker is to be the "weakest link." They are designed to *blow* or *trip* (disconnect) *before* a dangerous amount of current can melt your wires or start a fire.
- **A. Fuses:**
 - **How it works:** A simple device with a small wire inside. If the current is too high, the wire *melts* and breaks the circuit.
 - **Pros:** Very cheap.
 - **Cons: One-time use.** After it blows, you must throw it away and replace it.
- **B. Circuit Breakers (CB):**
 - **How it works:** A mechanical switch. If the current is too high, an internal mechanism (thermal or magnetic) *trips* the switch to the "OFF" position.
 - **Pros: Reusable.** You just need to find the fault (e.g., unplug the faulty appliance) and flip the switch back "ON."
 - **Cons:** More expensive than a fuse.
- **Where to install:** You *must* install breakers at key points:
 1. Between the solar panel array and the charge controller.

2. Between the charge controller and the battery bank.
3. Between the battery bank and the inverter (a large, heavy-duty breaker).

3. Protective Devices: Surge Arresters

- **Definition:** A Surge Protection Device (SPD) or "surge arrester."
- **Purpose:** To protect your expensive equipment (inverter, charge controller) from a "surge," which is a very short, very high spike in voltage.
- **Cause of Surges:** The most common cause is a **lightning strike** near the house.
- **How it works:** The SPD is connected between the "live" wires and the "ground" wire. It acts like a "gate." Normally, it does nothing. But when it "sees" a dangerous voltage spike (like from lightning), it instantly *opens the gate* and diverts the entire surge safely into the ground, *before* it can reach your inverter.

4. Proper Earthing/Grounding (Revisited)

- Grounding (Week 4) is not just for component protection; it is the **#1 personal safety device**.
- It protects *you* from **electric shock** by providing a safe path for fault currents.
- **Every metal part must be grounded:** Panel frames, mounting rails, inverter chassis.

5. Safe Handling of Tools and Batteries

- **Tools:** Use *insulated* tools (with rubber handles) when working on live circuits.
- **Batteries (Lead-Acid):**
 - **Acid:** Always wear safety glasses and gloves. Battery acid is corrosive and can blind you.
 - **Gas:** Charge batteries in a *ventilated* area. They release explosive hydrogen gas. Never smoke or create sparks near them.
 - **Short Circuit: NEVER** touch a metal tool (like a spanner) across the positive (+) and negative (-) terminals of a battery. It will create a massive spark, an explosion, and can weld the tool to the battery.

6. Practical: Installing a DC Circuit Breaker

- (Teacher demonstrates with a 12V battery, a 10A DC breaker, and a 12V bulb).
 1. Disconnect all power.
 2. Identify the "LINE" (from battery) and "LOAD" (to bulb) side of the breaker.
 3. Connect the wire from the **Positive (+) terminal of the battery** to the **LINE** terminal of the breaker.
 4. Connect the wire from the **LOAD** terminal of the breaker to the **positive (+) input of the 12V bulb**.
 5. Connect the negative (-) of the bulb *directly* to the negative (-) of the battery.
 6. Turn the breaker "ON." The bulb lights up.
 7. *Simulate a fault:* Create a (safe, low-current) short circuit. The breaker *trips* to "OFF" and the light goes out.

Evaluation:

1. What is the "Golden Rule" of electrical safety?
2. What is the main difference between a fuse and a circuit breaker?
3. What is the purpose of a "surge arrester"?
4. Why should you never touch a metal spanner across both terminals of a battery?
5. What does "overcurrent" mean, and what device protects against it?

Assignment:

1. Define: Circuit Breaker, Fuse, and Surge Arrester.
2. Why is it dangerous to charge lead-acid batteries in a sealed, unventilated room?
3. What is the main purpose of "grounding" (earthing) in a solar installation?
4. Draw a simple diagram showing where a circuit breaker should be placed between a battery and an inverter.
5. What does "PPE" stand for, and what 3 pieces of PPE should you use when working with batteries?

WEEK 8: MAINTENANCE AND SERVICING OF PV SYSTEMS



Content: (I) Preventive and corrective maintenance (II) Daily, weekly, and monthly schedules (III) Tools for maintenance (IV) Troubleshooting common issues (V) Group activity: Creating a maintenance checklist.

Performance Objectives: By the end of this lesson, students should be able to:

1. Differentiate between "Preventive" and "Corrective" maintenance.
2. Create a simple maintenance schedule (daily, monthly, yearly).
3. List the tools required for a basic maintenance check.
4. Diagnose and troubleshoot at least three (3) common solar system faults.
5. Develop a maintenance checklist for a residential solar installation.

Lesson Content:

1. Types of Maintenance

- **A. Preventive Maintenance (Proactive):**
 - **Definition:** Routine, scheduled checks and servicing you perform *before* a problem happens.
 - **Goal:** To prevent faults, maximize efficiency, and extend the life of the system.
 - **Example:** Cleaning solar panels so they don't get too dirty and lose power.
- **B. Corrective Maintenance (Reactive):**
 - **Definition:** Repair work you perform *after* a component has failed or a fault has occurred.
 - **Goal:** To fix a problem and get the system back online.
 - **Example:** The inverter has stopped working, so you must troubleshoot and replace it.
 - **Rule:** Good *preventive* maintenance reduces the need for *corrective* maintenance.

2. Sample Maintenance Schedule

- **Daily (User Check):**
 - Glance at the inverter/controller screen. Are there any red lights or error codes?
 - Is the system producing power?
- **Monthly (Owner Check):**
 - **Panel Cleaning:** If panels are in a dusty area, wash them (with water and a soft brush *in the early morning or evening* - never when hot).
 - **Battery Check (Lead-Acid):** Check the water level (if flooded) and clean the terminals of any white powder (corrosion).
 - **Look & Listen:** Check for any loose, hanging wires. Listen for any strange "buzzing" sounds from the inverter.
- **Yearly (Installer Check):**
 - **Professional Service:** An installer should visit once a year to:
 - Tighten *all* electrical connections (terminals) with a screwdriver/spanner.

- Test the voltage and current of each panel string.
- Check the health of the battery bank.
- Test the grounding (earthing) connection.

3. Tools for Maintenance

- Multimeter (for testing voltage and current)
- Insulated Screwdriver Set and Spanners (for tightening terminals)
- Soft Brush and Water (for panel cleaning)
- Battery Terminal Cleaner
- Safety PPE (Gloves, Glasses)

4. Troubleshooting Common Issues

- **Problem 1: "System is not charging batteries" (Daytime)**
 - **Possible Causes (in order):**
 1. The solar panel breaker is "OFF" (tripped). -> *Solution: Turn it on.*
 2. Heavy clouds or severe panel shading. -> *Solution: Wait for sun / cut tree branches.*
 3. Panel wiring fault (loose MC4 connector). -> *Solution: Test voltage at controller.*
 4. Charge controller is faulty. -> *Solution: Test and replace.*
- **Problem 2: "Inverter is beeping and has a red light"**
 - **Possible Causes:**
 1. **Overload:** The user plugged in too many appliances (e.g., an electric iron). -> *Solution: Unplug the heavy appliance and reset.*
 2. **Low Battery:** The batteries are empty (especially at night). -> *Solution: Turn off loads and wait for sun to recharge.*
 3. **High Temperature:** The inverter is overheating (fans blocked). -> *Solution: Clear dust/blockage, ensure ventilation.*
- **Problem 3: "Battery is not lasting long"**
 - **Possible Causes:**

1. The user is *using more power* than they think (overload).
2. Battery is old and has reached the end of its life.
3. A battery terminal is loose or corroded. -> *Solution: Clean and tighten.*
4. (Lead-Acid) Battery needs water.

5. Group Activity: Creating a Maintenance Checklist

- (Teacher guides students to create a simple checklist for a homeowner):
 - **Monthly Checklist:**
 - ☐ Check inverter screen for error codes. (Y/N)
 - ☐ Clean dust off solar panels. (Y/N)
 - ☐ Check battery terminals for corrosion. (Y/N)
 - ☐ Check battery water level (if applicable). (Y/N)
 - ☐ Look for any loose or damaged wires. (Y/N)

Evaluation:

1. What is the difference between preventive and corrective maintenance?
2. Why should you *never* clean a solar panel with harsh soap or in the hot afternoon sun?
3. Your inverter is beeping. What is the *most likely* cause? (Overload).
4. Your system stops working at night much faster than it used to. What is the *most likely* component to blame? (The battery).
5. What is the *first* thing you should check if the system isn't charging on a sunny day? (The circuit breaker).

Assignment:

1. Define Preventive and Corrective Maintenance.
2. Create your own 6-point "Monthly Maintenance Checklist" for a solar homeowner.
3. Why is it important to *tighten* electrical connections once a year? (Hint: A loose connection creates *resistance* and *heat*).
4. Your customer says their inverter is beeping and showing a "LOW BATT" warning. What do you tell them to do?

5. What safety tool is essential for checking if a circuit is "live" before you work on it?



WEEK 9: RENEWABLE ENERGY POLICIES AND REGULATIONS



Content: (I) Introduction to renewable energy policies in Nigeria (II) Agencies promoting solar (REA, NEMSA, NERC) (III) Standards and safety codes (IV) Discussion: Why regulations are important.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Policy" and "Regulation" in the context of energy.
2. Identify at least two (2) key government agencies involved in electricity in Nigeria.
3. Explain the primary role of the Rural Electrification Agency (REA).
4. Explain the primary role of NEMSA.
5. Discuss why standards and regulations are important for both installers and consumers.

Lesson Content:

1. What are Policies and Regulations?

- **Policy:** A "plan" or "goal" set by the government.
 - *Example:* "The Nigerian government has a *policy* to increase the use of renewable energy by 2030."
- **Regulation (or Law):** The specific *rules* and *standards* that must be followed to achieve that policy.
 - *Example:* "The government has a *regulation* that all imported solar panels must meet a minimum quality standard."

2. Key Government Agencies in Nigeria's Electricity Sector To install solar (especially hybrid/grid-tied), you must be aware of the "bosses" of electricity:

- **A. NERC (Nigerian Electricity Regulatory Commission):**
 - **Role:** The main "referee" for the *entire* electricity sector.
 - **Functions:** They set the rules for the industry. They issue licenses to DisCos (like Eko, Ikeja) and GenCos (like Egbin). They set the *electricity tariff* (the price you pay per unit).
- **B. REA (Rural Electrification Agency):**
 - **Role:** The agency specifically created to bring electricity to **rural and unserved communities**.
 - **Functions:** They manage government funds (and funds from the World Bank) to build "mini-grids" (small, local power stations, often using solar) in villages, markets (e.g., Sura Market solar), and universities ("Energizing Education" program).
 - **Importance for Solar:** The REA is one of the *biggest* promoters of solar energy in Nigeria.
- **C. NEMSA (Nigerian Electricity Management Services Agency):**
 - **Role:** The "police" of electrical installations.

- **Functions:** Their job is to **enforce standards, test equipment, and certify installers.**
- **Importance:** NEMSA's job is to make sure your installation is *safe*. They test new meters, inspect installations, and can *disconnect* a building if it is wired unsafely. A good installer should follow NEMSA's codes.
- **D. SON (Standards Organisation of Nigeria):**
 - **Role:** Sets the minimum quality standards for *all* products sold in Nigeria, including solar components.
 - **Function:** They are supposed to stop fake or "sub-standard" panels and batteries from entering the country.

3. Standards and Safety Codes

- **Code:** A "rulebook" for safe installation. (e.g., NEMSA's guidelines).
- **Examples of Standards/Codes:**
 - **Cable Sizing:** "You must use a 6mm cable for this current."
 - **Grounding:** "The earthing rod must be at least 1.5 meters deep."
 - **Labeling:** "All circuit breakers must be clearly labeled."

4. Discussion: Why are Regulations Important?

- **A. For the Customer (Consumer Protection):**
 - Ensures they are not cheated with "fake" or "sub-standard" products (e.g., a panel labeled 300W that is only 200W).
 - Guarantees that the installation is *safe* and will not cause a fire or electrocute them.
- **B. For the Installer (Professionalism):**
 - Creates a professional industry by separating the qualified, *certified* installers from the untrained, "quack" installers.
 - Gives the installer clear rules to follow, which protects them from being blamed if an accident happens.
- **C. For the Nation (Grid Stability):**

- For hybrid/grid-tied systems, regulations are *critical*. They prevent a badly installed system from "back-feeding" power and *electrocuting* a "NEPA" worker who is trying to fix a line.

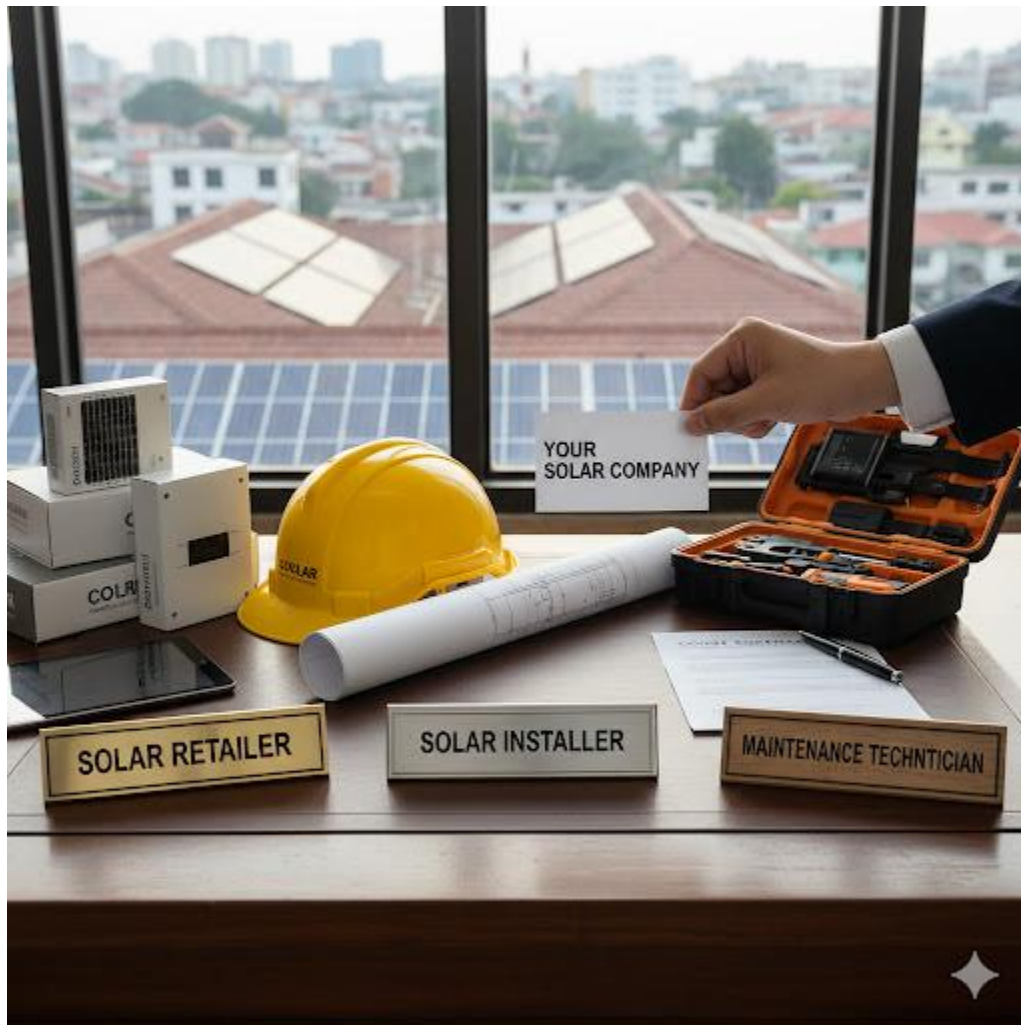
Evaluation:

1. What is the difference between a "policy" and a "regulation"?
2. Which agency is the main "referee" that sets electricity prices? (NERC).
3. Which agency is focused on building solar "mini-grids" in villages? (REA).
4. Which agency is the "police" that inspects installations for safety? (NEMSA).
5. Give one reason why regulations are *good* for the customer.

Assignment:

1. Define Policy and Regulation.
2. Write the full meaning of these acronyms: NERC, REA, and NEMSA.
3. In your own words, what is the main job of the REA?
4. Why is it important for the Standards Organisation of Nigeria (SON) to check imported solar panels?
5. (Opinion) Why do you think so many "quack" or untrained installers exist in Nigeria, and what could be done about it?

WEEK 10: ENTREPRENEURSHIP IN SOLAR ENERGY (REVISITED)



Content: (I) Business setup (retail, installation, maintenance) (II) Customer relations and branding (III) Cost estimation and quotation writing (IV) Case study (V) Group presentation.

Performance Objectives: By the end of this lesson, students should be able to:

1. Identify three (3) distinct business opportunities in the solar sector.
2. Define "branding" and its importance for an installer.
3. Differentiate between "cost" and "price."
4. Write a simple, professional quotation for a basic solar installation.

5. Develop and present a simple "solar business plan."

Lesson Content:

1. Business Opportunities in Solar (Recap) As an entrepreneur, you don't just have to be an installer. You can specialize:

- **A. Solar Retailer (Sales):**
 - **Business:** You run a shop (physical or online) that *sells* components (panels, batteries, etc.) to the public and to other installers.
 - **Skill:** Good at marketing, sales, and managing inventory.
- **B. Solar Installer (Service):**
 - **Business:** You *design* and *install* complete systems for clients. This is the most common route.
 - **Skill:** Strong technical knowledge (sizing, wiring), problem-solving.
- **C. Solar Maintenance (Service):**
 - **Business:** You offer "maintenance contracts" to homes and businesses that *already* have solar. You visit every 6 months to clean panels, test batteries, and service the system for a fee.
 - **Skill:** Good at troubleshooting, customer relations, reliable.

2. Customer Relations and Branding

- **Branding:** Your "brand" is your *reputation*. In the solar industry, your brand must be built on **TRUST**.
- **How to Build Trust:**
 - **Professionalism:** Show up on time. Wear a clean uniform/t-shirt with your company logo.
 - **Honesty:** **Never** sell fake products. **Never** lie about a component's capacity. A bad reputation spreads fast.
 - **Good Work:** Do a neat, safe installation (good cable management). A clean-looking installation gives the customer confidence.

- **Customer Service:** Explain to the customer *how* their system works. Show them what the lights mean, what to do in a fault, and give them your maintenance checklist (from Week 8).

3. Cost Estimation and Quotation Writing This is a vital business skill.

- **A. Cost Estimation (Your Cost):**
 - This is how much the job will *cost you* as the installer.
 - Total Cost = Cost of Components + Cost of Labor + Cost of Transport
- **B. Quotation (The Price):**
 - This is the *formal document* you give to the customer *before* the job, showing the *price* they will pay.
 - Price = Total Cost + Your Profit (Markup)
- **How to Write a Simple Quotation (Example):**

YOUR SOLAR COMPANY (Your Name/Logo/Phone)
To: [Customer Name]
Date: [Date]
Subject: Quotation for 1kW Solar System
Item Description
320W Monocrystalline Panel
12V 200Ah Lead-Acid Battery
1.5kW Pure Sine Wave Inverter
40A MPPT Charge Controller
Wires, Breakers, Rails, Lugs
Installation & Labor Fee
TOTAL
<i>Terms: 50% advance payment, 50% on completion.</i>

4. Case Study: Successful Nigerian Solar Startups

- **Discussion:** Teacher leads a discussion on real Nigerian solar companies (e.g., *Arnergy*, *Lumos*, *Rensource*). How do they market themselves? What services do they offer?

5. Group Presentation: "My Solar Business Plan"

- **Task:** Students get into groups.
- **Presentation:** Each group presents a 2-minute "pitch" for their new solar business. They must state:
 1. The *Name* of their company.
 2. Their *Slogan* (e.g., "Reliable Power, Honest Service").
 3. Their *Business Type* (Retail, Installation, or Maintenance).
 4. *Why* customers should choose them (their brand promise).

Evaluation:

1. List the three main business types in the solar industry.
2. Why is "trust" the most important part of a solar installer's brand?
3. What is the difference between your "cost" and the customer's "price"?
4. What is a "Quotation"?
5. What is the largest cost item in the sample quotation? (The batteries).

Assignment:

1. Create a name and a simple slogan for your own (imaginary) solar installation business.
2. What is the first rule of good customer relations as an installer?
3. Using the sample as a guide, write a simple **Quotation** for the small DC system you designed in Week 3 (12V, 7W x 3 bulbs).
 - (Invent prices for: 1 x 50W panel, 1 x 30Ah battery, 1 x 10A PWM controller, 3 x 7W DC bulbs, Wires, and Labor).
4. What does "Markup" (or "Profit") mean?
5. Why is it important to write down "Payment Terms" on a quotation?

WEEK 11: REVISION

Topic: Revision

Content: Comprehensive review of all topics covered.

Note: This week is dedicated to a comprehensive review of all topics covered in the First Term, especially in preparation for the end-of-term examination.

Revision Checklist:

1. JSS 2 Review & Configs (Week 1):

- What are the 4 main components?
- What is a Standalone vs. Grid-Tied vs. Hybrid system?
- What is the difference between MPPT and Li-ion (new tech) vs. PWM and Lead-Acid (old tech)?

2. Advanced Components (Week 2):

- Mono vs. Poly panels (appearance, efficiency).
- Li-ion vs. Lead-Acid (DoD, lifespan, weight, cost).
- MPPT vs. PWM (efficiency, "harvesting" power).
- Pure Sine Wave vs. Modified Sine Wave (safety for appliances).

3. System Sizing (Week 3):

- What is the difference between Power (W) and Energy (Wh)?
- Formula: $\text{Energy (Wh)} = \text{Power (W)} \times \text{Time (h)}$
- How to size an Inverter (Total W + safety factor).
- How to size a Battery ($\text{Total Wh} / (\text{V} \times \text{DoD})$).
- How to size a Panel Array ($\text{Total Wh} / (\text{PSH} \times \text{Efficiency})$).

4. Installation (Week 4):

- Tools (Multimeter, Crimper, Stripper).
- Mounting (Face South, 10-15 degree tilt).
- **Series Connection:** (+ to -) -> **Voltage adds**, Amps are same.
- **Parallel Connection:** (+ to +) -> **Amps add**, Voltage is same.
- Grounding (Safety for electrocution).

5. Safety Devices (Week 6):

- Fuse vs. Circuit Breaker (Overcurrent protection, one-time vs. reusable).
- Surge Arrester (Lightning protection).
- Battery Safety (Acid, Gas, Short circuits).

6. Maintenance (Week 8):

- Preventive (before) vs. Corrective (after).
- Schedule (Daily checks, Monthly cleaning, Yearly tightening).
- Troubleshooting (Overload, Low Battery, Panel Shading, Breaker Trip).

7. Regulations (Week 9):

- NERC (Regulator, sets tariffs).
- REA (Rural/village projects, mini-grids).
- NEMSA (Safety police, standards, certification).

8. Entrepreneurship (Week 10):

- Business types (Retail, Install, Maintenance).
- Branding (Trust, Honesty).
- Cost (your cost) vs. Price (customer price).
- How to write a Quotation.

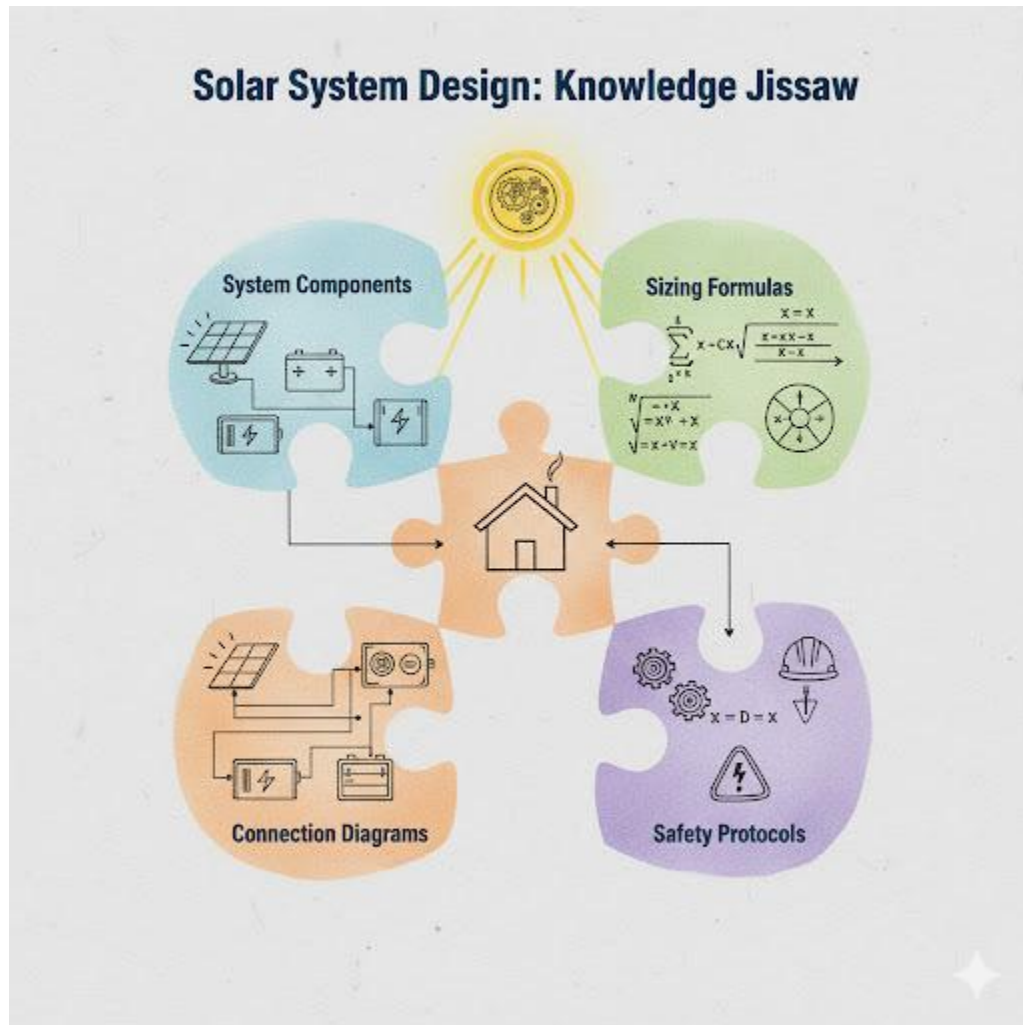
Activity:

- Comprehensive Q&A session.
- Review of past exam questions.
- Final review of the "My Solar Business Plan" presentations.

**SECOND TERM - JSS 3 SOLAR PHOTOVOLTAIC (PV) INSTALLATION AND
MAINTENANCE LESSON NOTES**



WEEK 1: REVISION OF FIRST TERM WORK



Content: (I) Recap of advanced system components and installation (II) Review of system design and sizing (III) Quick diagnostic test and discussion on students' practical experiences.

Performance Objectives: By the end of this lesson, students should be able to:

1. Recall and explain the differences between advanced components (Mono/Poly, Li-ion/Lead-Acid, MPPT/PWM, PSW/Modified).
2. Summarize the complete process of system sizing, from load calculation to panel array sizing.
3. Explain the difference between a series and a parallel connection and their mathematical effects.

4. Recap the key safety devices (breakers, fuses, grounding) and their functions.
5. Discuss practical challenges faced during JSS 3 First Term activities and their solutions.

Lesson Content:

1. Comprehensive Recap of JSS 3 First Term This week is a vital review to consolidate the advanced concepts learned in the first term before proceeding.

- **A. Advanced Components (Review):**

- **Panels:** Monocrystalline (black, cut corners, high efficiency) vs. Polycrystalline (blue, square cells, lower cost).
- **Batteries:** Lead-Acid (heavy, 50% DoD, 3-5yr life) vs. Lithium-ion (lightweight, 80-100% DoD, 10-15yr life).
- **Controllers:** PWM (cheaper, less efficient, for small systems) vs. MPPT (smart, 15-30% more power harvest, for large systems).
- **Inverters:** Modified Sine Wave (cheap, "blocky" wave, bad for electronics) vs. Pure Sine Wave (PSW) (clean power, safe for all appliances).

- **B. System Design and Sizing (Review):**

- **Step 1: Load Estimation.** The most important step. Energy (Wh) = Power (W) x Time (h). Calculate Total Daily Energy (Wh).
- **Step 2: Inverter Sizing.** Based on total *simultaneous* power in Watts. Inverter Size (W) = Total Watts x 1.5 (safety factor).
- **Step 3: Battery Sizing.** Based on total *energy* in Watt-hours. Battery Ah = (Total Wh x Days of Autonomy) / (System Voltage x DoD).
- **Step 4: Panel Array Sizing.** Must recharge the daily energy used. Panel Array (W) = Total Wh / (Peak Sun Hours x System Efficiency).

- **C. Installation & Safety (Review):**

- **Connections:**
 - **Series (+ to -):** Voltage adds up, Current stays the same. (e.g., 2 x 12V = 24V).
 - **Parallel (+ to +, - to -):** Current adds up, Voltage stays the same. (e.g., 2 x 100Ah = 200Ah).

- **Safety Devices:**

- **Circuit Breakers/Fuses:** Protect wires from *overcurrent* and fire.
- **Surge Arresters:** Protect equipment from *overvoltage* (lightning).
- **Grounding (Earthing):** Protects *people* from *electric shock*.

2. Quick Diagnostic Test & Discussion

- **Diagnostic Test (Written or Oral):**

1. What is the main difference between MPPT and PWM?
2. What is the main danger of using a modified sine wave inverter?
3. You need a 24V, 200Ah battery bank. You have 12V, 100Ah batteries. How many do you need and how do you connect them? (Answer: 4 batteries total. Two sets of two in series, and then connect those two sets in parallel).
4. What is the *first* step in designing *any* solar system? (Load calculation).
5. What is the purpose of grounding?

- **Class Discussion:**

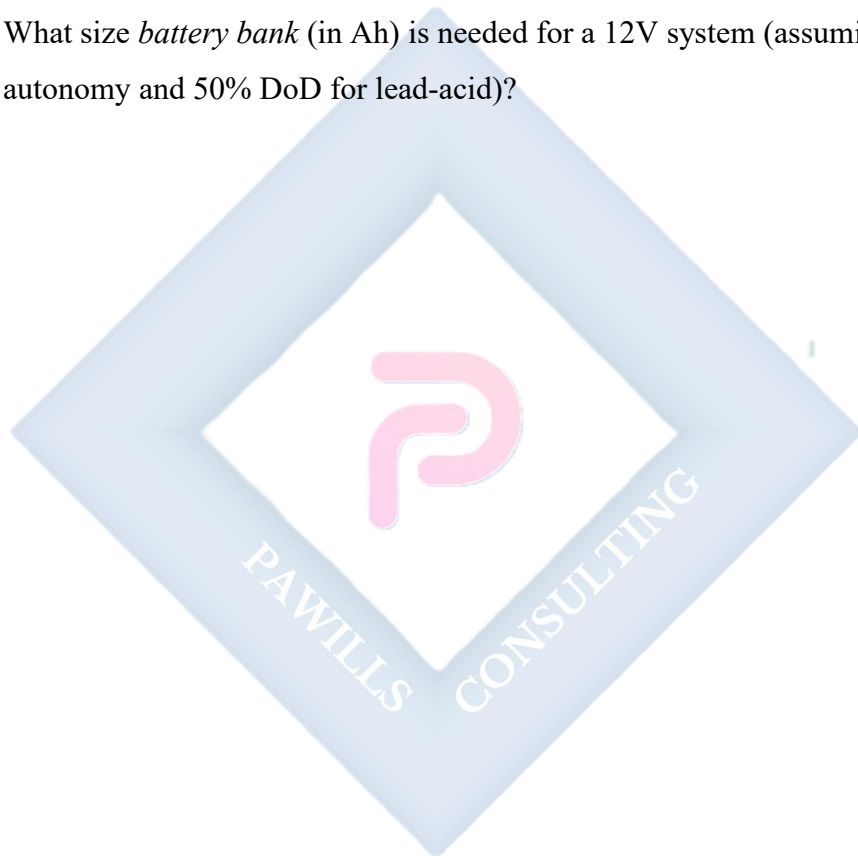
- "What was the most difficult part of the system sizing calculation for you?"
- "Did anyone have trouble with the series/parallel connection practical?"
- "Based on last term, what is the most common mistake an installer can make?"

Evaluation:

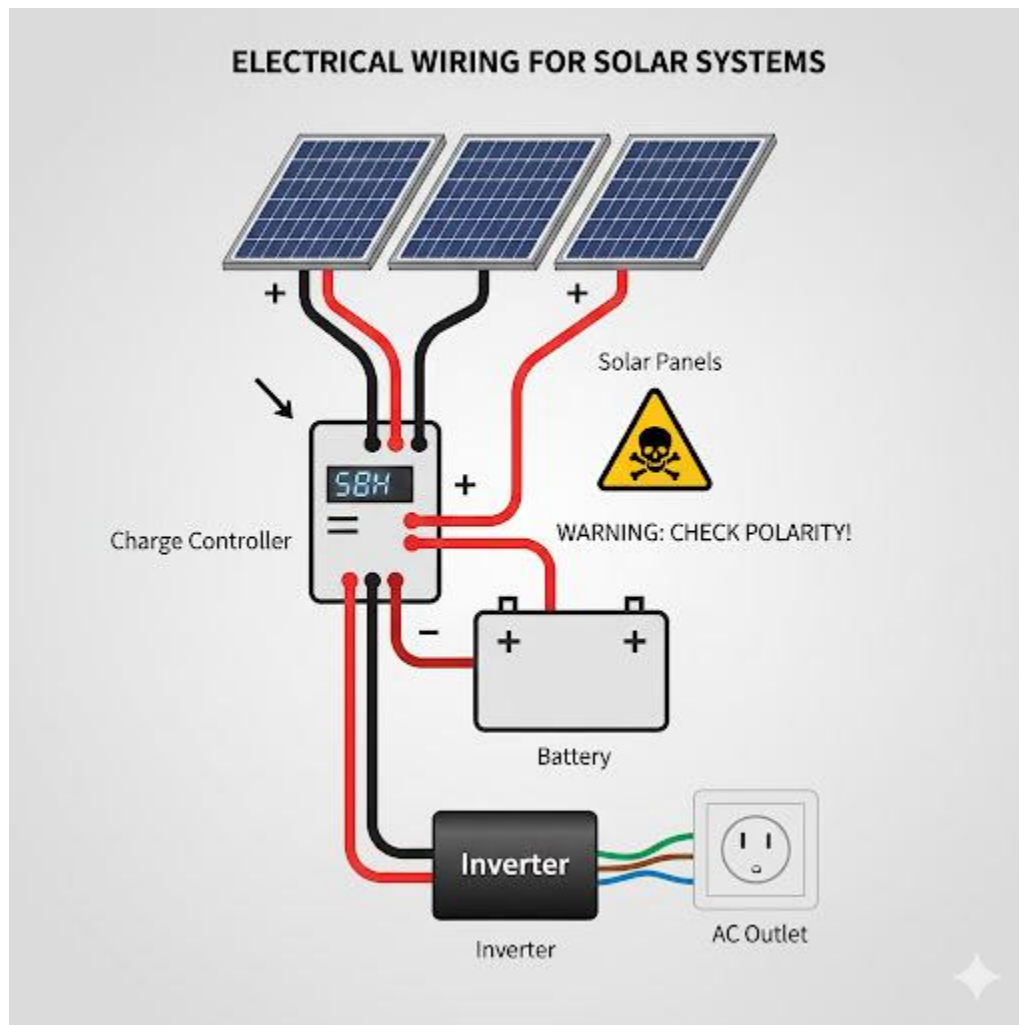
1. Explain the difference between a Monocrystalline and a Polycrystalline panel.
2. A customer wants to power a N1,000,000 TV and a medical device. What type of inverter *must* you use and why?
3. What are the two main benefits of a Lithium-ion battery over a Lead-Acid battery?
4. You have two 12V, 150Ah batteries. If you connect them in *series*, what is the final voltage and capacity?
5. What is the main function of a circuit breaker?

Assignment:

1. In your own words, describe the complete 4-step process for sizing a standalone solar system.
2. Draw the wiring diagram for connecting four 12V, 100Ah batteries to create a 24V, 200Ah bank.
3. Why is an MPPT charge controller more efficient than a PWM controller?
4. List three common faults or maintenance issues a solar installer must check for.
5. A client's load is 500W and their daily energy is 2,000Wh.
 - What size *inverter* should you recommend (with a 1.5 safety factor)?
 - What size *battery bank* (in Ah) is needed for a 12V system (assuming 1 day autonomy and 50% DoD for lead-acid)?



WEEK 2: ELECTRICAL WIRING FOR SOLAR SYSTEMS



Content: (I) Introduction to solar electrical wiring concepts (II) Types of electrical cables (AC/DC cables) (III) Colour coding and polarity awareness (IV) Cable sizing and voltage drop (V) Practical: Connecting components correctly.

Performance Objectives: By the end of this lesson, students should be able to:

1. Differentiate between AC and DC cables and their applications in a solar system.
2. State the standard IEC (international) colour codes for AC and DC wiring.
3. Define "Polarity" and explain the danger of reversing it.
4. Define "Cable Sizing" and "Voltage Drop" and explain why they are important.
5. Demonstrate the correct and safe connection of panels, controllers, and batteries.

Lesson Content:

1. Introduction to Solar Electrical Wiring

- Wiring is the "circulatory system" of the solar installation. Poor wiring is the **#1 cause of system failure and fires**.
- A solar system has two distinct types of wiring:
 - **DC (Direct Current) Side:** The "solar side." This includes all wiring *between* the panels, charge controller, and batteries.
 - **AC (Alternating Current) Side:** The "appliance side." This is the wiring *after* the inverter, which goes to your wall sockets, lights, and appliances.

2. Types of Electrical Cables

- **A. DC Cables (PV Cables):**
 - **Description:** These are special, single-core cables designed specifically for solar panels. They are **UV-resistant** (the insulation won't crack in the sun) and high-voltage (rated for 1000V+).
 - **Use:** *Only* for connecting the solar panels, from the roof down to the charge controller.
- **B. DC Battery Cables:**
 - **Description:** Thick, multi-stranded cables (like "welding cables") designed to carry very high *current* (Amps) at low voltage (12V, 24V, 48V).
 - **Use:** For connecting the battery bank to the inverter and the charge controller.
- **C. AC Cables (e.g., TW/THHN):**
 - **Description:** Standard house wiring cables (e.g., 1.5mm, 2.5mm, 4mm).
 - **Use:** For wiring the *output* of the inverter to the house's AC distribution board and sockets.

3. Colour Coding and Polarity Awareness

- **Polarity:** The direction of electrical flow. DC power has a fixed **Positive (+)** and **Negative (-)**.

- **DANGER: Reversing polarity** (connecting + to - and - to +) will **instantly destroy** your equipment, especially the charge controller and inverter. This is an expensive mistake.
- **Standard Colour Codes:**
 - **DC Wiring (Solar/Battery):**
 - **Positive (+): RED**
 - **Negative (-): BLACK**
 - **AC Wiring (Inverter Output - IEC Standard):**
 - **Live (L): BROWN**
 - **Neutral (N): BLUE**
 - **Earth (E/Ground): GREEN/YELLOW stripes**
 - *Note:* Nigeria sometimes uses old British colours (Red for Live, Black for Neutral). This can be confusing. As a solar professional, it's best to stick to the modern IEC standard.

4. Cable Sizing and Voltage Drop

- **Cable Sizing:** The process of choosing the correct *thickness* (measured in mm² or AWG) for a cable.
 - **Why?** A wire is like a water pipe. A small pipe can only carry a small amount of water. A thin wire can only carry a small amount of *current (Amps)*.
 - **If a cable is too thin:** It will **overheat, melt, and cause a fire.**
 - **Rule of Thumb:** High-current paths (like battery-to-inverter) need *very thick* cables. Low-current paths (like panel-to-controller) can be thinner.
- **Voltage Drop:**
 - **Definition:** The small amount of voltage that is *lost* as electricity travels down a long wire.
 - **Why it matters:** If your solar panels are 50 meters away from your controller and you use a thin wire, a 12V panel might only deliver 10V. This is *not enough* to charge the battery.
 - **Solution:**
 1. Keep cable runs (distances) as *short* as possible.

2. Use a *thicker* cable for long distances.
3. Design a *higher voltage* system (e.g., 24V or 48V) because voltage drop is less of a problem at higher voltages.

5. Practical: Connecting Components

- (Teacher demonstrates the correct wiring order for a system).
- **ORDER OF CONNECTION (CRITICAL):**
 1. **Step 1:** Connect the **Charge Controller** to the **Battery Bank**. (Controller needs to "see" the battery voltage *first*).
 2. **Step 2:** Connect the **Solar Panels** to the **Charge Controller**. (You should see the charging light come on).
 3. **Step 3:** Connect the **Inverter** directly to the **Battery Bank** (using its thick cables).
 4. **Step 4:** Connect the **AC Loads** (e.g., a fan) to the **Inverter**.
- **ORDER OF DISCONNECTION (Opposite):**
 1. Disconnect AC Loads.
 2. Disconnect Inverter.
 3. Disconnect Solar Panels.
 4. Disconnect Battery.

Evaluation:

1. What is the main difference between a "DC cable" and an "AC cable" in a solar system?
2. What is the standard DC colour code for Positive and Negative?
3. What is "polarity," and what happens if you reverse it on a charge controller?
4. What is "cable sizing," and what is the danger of using a cable that is too thin?
5. What is the very *first* component you should connect to a charge controller?

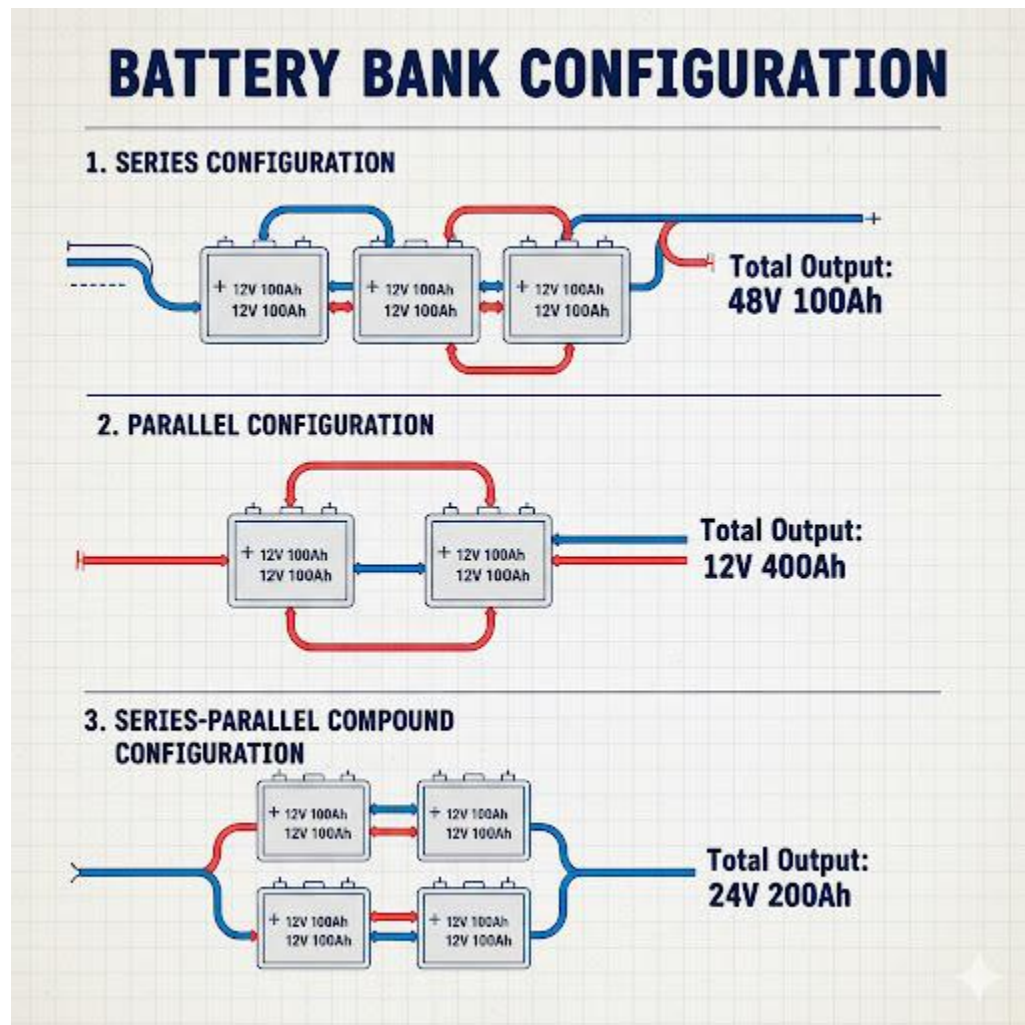
Assignment:

1. Define: Polarity, Cable Sizing, and Voltage Drop.
2. Draw a simple diagram showing the 4 main components (Panel, Controller, Battery, Inverter) and the AC/DC loads, labeling the wiring as either "AC" or "DC."

3. Why is a special "PV cable" used for solar panels instead of regular indoor wire?
4. What is the correct *order of disconnection* for a solar system?
5. A customer wants to place their panels 100 meters away from their batteries. What problem will they face, and what are two possible solutions?



WEEK 3: BATTERY BANK CONFIGURATION



Content: (I) Series and parallel connections in batteries (II) Battery capacity and voltage calculation (III) Factors affecting battery performance and lifespan (IV) Maintenance and storage of batteries (V) Practical: Building a 24V battery bank.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Battery Bank."
2. Calculate the final voltage and capacity (Ah) for a *series* connection.
3. Calculate the final voltage and capacity (Ah) for a *parallel* connection.
4. Assemble a 24V battery bank from 12V batteries.
5. List three factors that reduce a battery's performance and lifespan.

Lesson Content:

1. What is a Battery Bank?

- **Definition:** A "battery bank" is a group of two or more individual batteries connected together to function as a single, larger battery.
- **Purpose:** To get the required **Voltage (V)** (e.g., 24V or 48V) and **Capacity (Ah)** for a solar system.
- **Key Components:**
 - **Batteries:** The individual units (e.g., 12V, 200Ah).
 - **Battery Cables (Interconnects):** Short, *very thick* cables used to link the batteries.

2. Battery Connection Techniques (Recap & Advanced)

- **A. Series Connection (Building Voltage)**
 - **How:** Connecting batteries in a "chain." **Positive (+) to Negative (-).**
 - **Effect:**
 - **Voltage (V) ADDS UP.**
 - **Capacity (Ah) STAYS THE SAME.**
 - **Example 1:** Two 12V 100Ah batteries in Series.
 - $V_{\text{final}} = 12V + 12V = 24V$
 - $Ah_{\text{final}} = 100Ah$ (stays the same)
 - **Result:** A 24V, 100Ah bank.
 - **Rule:** You *must* use batteries of the **same capacity (Ah)**.
- **B. Parallel Connection (Building Capacity)**
 - **How:** Connecting batteries "side-by-side." **Positive (+) to Positive (+)** and **Negative (-) to Negative (-).**
 - **Effect:**
 - **Voltage (V) STAYS THE SAME.**
 - **Capacity (Ah) ADDS UP.**

- **Example 2:** Two 12V 100Ah batteries in Parallel.
 - $V_{\text{final}} = 12\text{V}$ (stays the same)
 - $Ah_{\text{final}} = 100\text{Ah} + 100\text{Ah} = 200\text{Ah}$
 - **Result:** A 12V, 200Ah bank.
 - **Rule:** You *must* use batteries of the **same voltage (V)**.
 - **C. Compound Connection (Series-Parallel)**
 - **How:** Combining both methods.
 - **Example 3:** Four 12V 100Ah batteries to make a **24V, 200Ah bank**.
1. **Step 1 (Series):** Create two *strings* of 24V.
 - (Batt 1 + Batt 2) in Series = 24V, 100Ah string.
 - (Batt 3 + Batt 4) in Series = 24V, 100Ah string.
 2. **Step 2 (Parallel):** Connect the two strings in *parallel*.
 - (String 1 + String 2) in Parallel = 24V, $(100\text{Ah} + 100\text{Ah}) = 200\text{Ah}$.
- **This is the standard method for building large, professional battery banks.**

3. Practical: Building a 24V Battery Bank

- **Goal:** Create a 24V bank from two 12V batteries.
- **Tools:** 2 x 12V batteries, 1 short battery interconnect cable, spanners.
- **Steps:**
 1. **Safety:** Wear gloves and safety glasses.
 2. **Identify Terminals:** Place the batteries side-by-side.
 3. **Make Series Connection:** Securely connect one cable from the **Positive (+) terminal of Battery 1** to the **Negative (-) terminal of Battery 2**.
 4. **Identify Bank Terminals:** The *remaining* terminals are now the terminals for the whole bank.
 - Bank Positive = **Negative (-) of Battery 1** (This is an error, should be Positive of Batt 1)
 - Bank Negative = **Positive (+) of Battery 2** (This is an error, should be Negative of Batt 2)

- **Correction:**

- **Bank Positive (+):** The open **Positive (+) terminal on Battery 1.**
- **Bank Negative (-):** The open **Negative (-) terminal on Battery 2.**

5. **Test:** Use a multimeter to measure the voltage between the final Bank Positive and Bank Negative. It should read ~24V-26V.

4. Factors Affecting Battery Performance and Lifespan

1. **Temperature (The #1 Killer):** Batteries are chemicals. High heat (like in Nigeria) *kills* them. A battery that lasts 5 years in Europe may only last 2 years here. They must be stored in a *cool, ventilated* place.
2. **Depth of Discharge (DoD):** Discharging a lead-acid battery 100% *will* damage it. A battery discharged to 50% daily will last *much longer* (e.g., 1000 cycles) than one discharged to 80% daily (e.g., 300 cycles).
3. **Charging Rate:** Charging too fast (with an oversized controller) or too slow (with undersized panels) can damage the battery.
4. **Maintenance (Lead-Acid):** If the water level in a flooded battery drops below the plates, the plates are exposed to air, oxidize, and are permanently damaged.

Evaluation:

1. What is a "battery bank"?
2. You connect two 12V, 200Ah batteries in **parallel**. What is the final voltage and capacity?
3. You connect two 12V, 200Ah batteries in **series**. What is the final voltage and capacity?
4. What is the *most damaging* environmental factor for a battery?
5. What does "DoD" stand for, and why is a 50% DoD better for a battery's life than an 80% DoD?

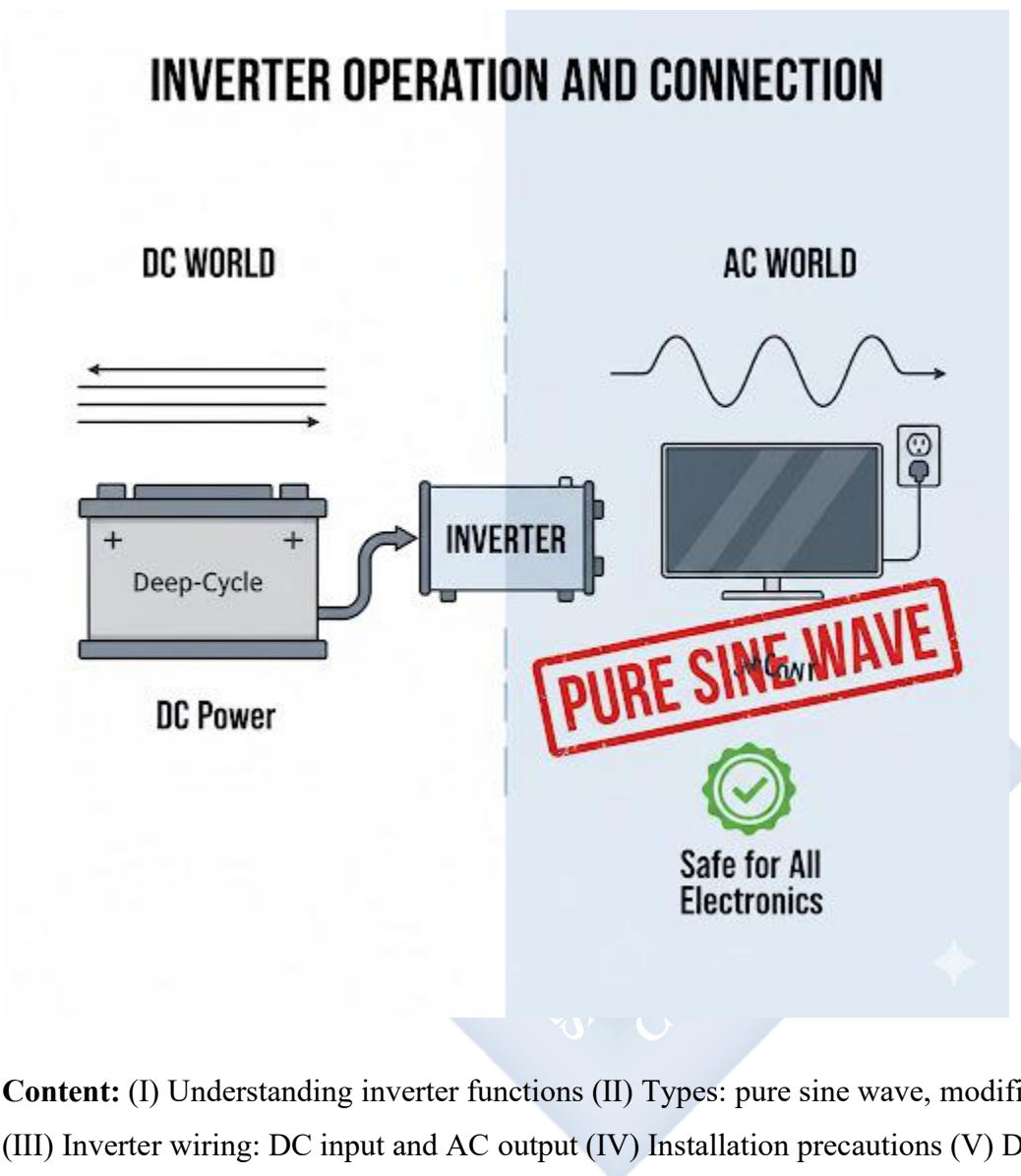
Assignment:

1. Define Series Connection and Parallel Connection (for batteries).

2. Draw a clear diagram of four 12V, 200Ah batteries connected to make a **48V, 200Ah** bank.
3. Draw a clear diagram of four 12V, 200Ah batteries to make a **12V, 800Ah** bank.
4. Draw a clear diagram of four 12V, 200Ah batteries to make a **24V, 400Ah** bank. (This is a compound connection).
5. List three essential maintenance tips for a lead-acid battery bank.



WEEK 4: INVERTER OPERATION AND CONNECTION



Content: (I) Understanding inverter functions (II) Types: pure sine wave, modified sine wave (III) Inverter wiring: DC input and AC output (IV) Installation precautions (V) Demonstration: Connecting an inverter.

Performance Objectives: By the end of this lesson, students should be able to:

1. State the primary function of an inverter (DC to AC conversion).
2. Re-explain the critical difference between Pure Sine Wave (PSW) and Modified Sine Wave (MSW) inverters.
3. Identify the DC input terminals and the AC output terminals on an inverter.
4. List three (3) critical precautions for installing an inverter.

5. Safely demonstrate connecting an inverter to a battery and a load.

Lesson Content:

1. Inverter Function (Recap)

- **Primary Function:** The inverter is the "heart" of an AC solar system. Its main job is to **invert** (change) **low-voltage Direct Current (DC)** from the battery bank into **high-voltage Alternating Current (AC)**.
 - *Example:* 12V DC → 230V AC
- **Secondary Functions (in modern inverters):**
 - **Battery Charger:** Many inverters (called "Inverter-Chargers") also have an AC input plug. This allows them to *charge* the batteries using grid (NEPA) power or a generator when solar is not enough.
 - **Automatic Transfer Switch (ATS):** Automatically switches from grid power to battery power (in milliseconds) when the grid fails, so your appliances don't turn off.

2. Types of Inverters (Recap)

- **A. Modified Sine Wave (MSW):**
 - **Output:** A "blocky," stepped wave. It is a "dirty" or "choppy" approximation of AC power.
 - **Use:** *Only* for simple loads like light bulbs or basic heaters.
 - **DANGER: Will damage** or destroy sensitive electronics like laptops, fridges, smart TVs, and medical equipment. It makes fans and motors buzz and overheat.
- **B. Pure Sine Wave (PSW):**
 - **Output:** A smooth, clean electrical wave that is identical to (or better than) grid power.
 - **Use: Safe for ALL appliances.** This is the *only* type that should be used for a professional home or office installation.

3. Inverter Wiring: Inputs and Outputs

An inverter has two distinct sides:

- **A. DC Input Side:**

- **Description:** This is the "low voltage, high current" side.
- **Terminals:** Two large, threaded terminals (posts), clearly labeled:
 - **POSITIVE (+):** Coloured **RED**.
 - **NEGATIVE (-):** Coloured **BLACK**.
- **Connection:** This side connects *directly* to the main battery bank terminals, using *very thick* battery cables.

- **B. AC Output Side:**

- **Description:** This is the "high voltage, low current" side.
- **Terminals:** This can be:

1. Standard wall sockets (for small inverters).
2. A "terminal block" (for large inverters) labeled **L** (Live), **N** (Neutral), and **E** (Earth/Ground).

- **Connection:** This side connects to the building's electrical distribution board (DB) or directly to the appliances.

4. Installation Precautions (Critical Safety)

1. **NEVER Reverse Polarity:** Connecting the battery's (+) to the inverter's (-) and vice-versa will **instantly destroy the inverter**. It will cause a large spark, blow the internal fuses, and void the warranty. **Always check: RED to RED, BLACK to BLACK.**
2. **Use THICK, SHORT Cables:** The DC (battery) side uses *extremely high current* (e.g., a 1000W inverter on 12V pulls over 80 Amps!). The cables *must* be very thick (as specified in the manual) and as *short as possible* (e.g., < 1.5 meters) to prevent overheating and fire.
3. **Install a Fuse/Breaker:** A large fuse or DC circuit breaker *must* be installed on the **Positive (+) battery cable** as close to the battery as possible. This protects the inverter and wires from a catastrophic short circuit.
4. **Ventilation:** Inverters get hot. They must be installed in a *cool, dry, and ventilated* area. Never cover the fans or install in a small, sealed cupboard.
5. **Grounding:** The inverter's metal chassis (body) *must* be connected to the system's grounding rod (Week 4).

5. Demonstration: Connecting an Inverter

- **Goal:** Safely power an AC fan from a 12V battery.
- **Tools:** 12V battery, 300W PSW inverter, 1 AC fan, correct DC cables, safety glasses.
- **Steps:**
 1. Place inverter and battery close together.
 2. Ensure inverter switch is **OFF**.
 3. **Connect Negative Cable:** Securely connect the BLACK cable to the inverter's (-) terminal, then to the battery's (-) terminal.
 4. **Connect Positive Cable:** Securely connect the RED cable to the inverter's (+) terminal.
 5. **FINAL STEP:** *Carefully* tap, then connect, the other end of the RED cable to the battery's (+) terminal. (A small, normal spark may occur).
 6. **Turn Inverter ON:** The inverter's "power" light should come on.
 7. **Test Load:** Plug the AC fan into the inverter's AC output socket and turn it on.
 8. **Test Disconnection:** Turn off the fan, turn off the inverter, disconnect the POSITIVE cable first.

Evaluation:

1. What are the *two* main functions of a modern inverter-charger?
2. What is the main danger of using a *Modified* Sine Wave inverter?
3. What are the DC input terminals on an inverter, and what colours are they?
4. What is the *most common* and *most expensive* mistake an installer can make when connecting an inverter to a battery?
5. Why must the battery cables be very *thick* and very *short*?

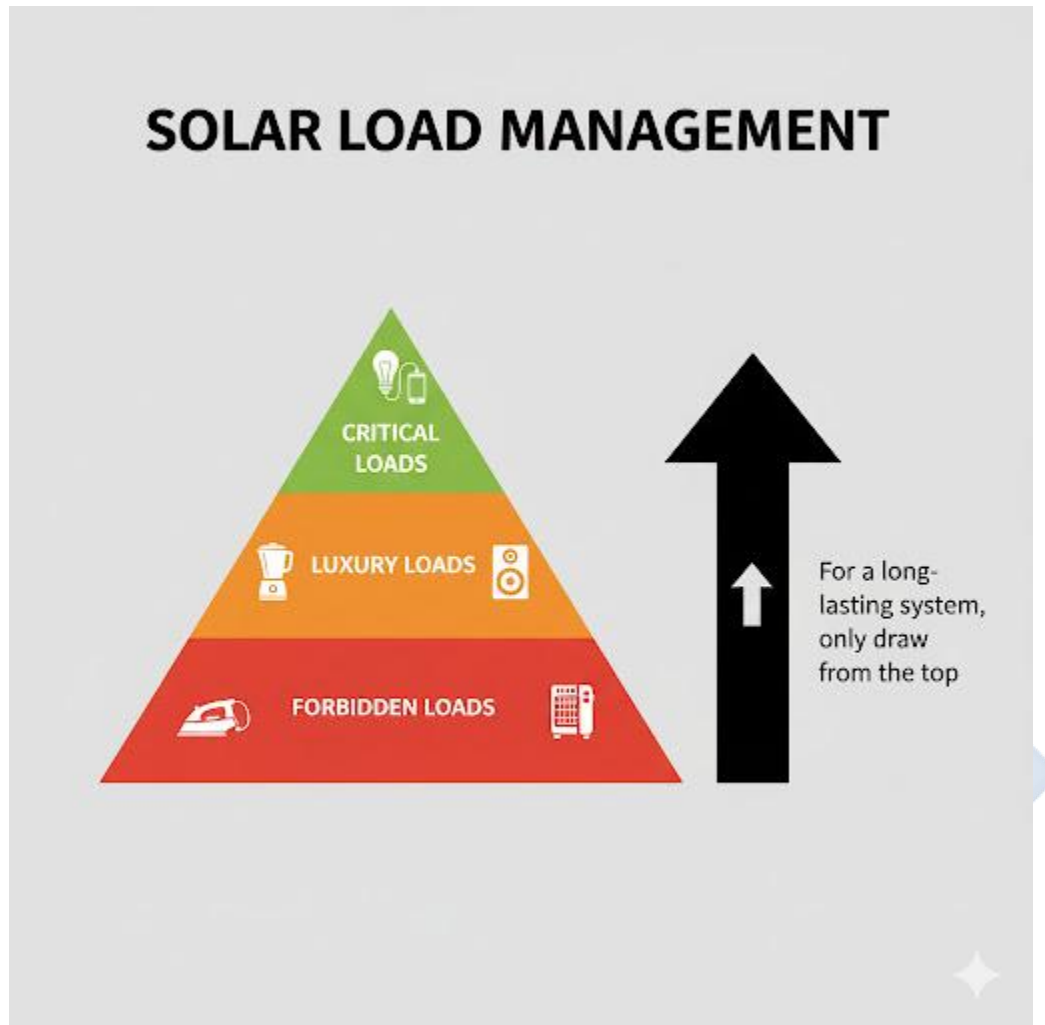
Assignment:

1. Define "Inverter" and "Pure Sine Wave."
2. Draw a simple diagram showing an inverter connected to a battery and an AC load. Label the "DC Input" and "AC Output."
3. List three (3) critical safety precautions to take when installing an inverter.

4. Why must an inverter be installed in a well-ventilated area?
5. A 1200W inverter is connected to a 12V battery. The current (Amps) it draws is $I = P / V$.
 - What is the *maximum current* it will draw?
 - Why would a 2.5mm AC (house) wire be *extremely dangerous* to use for this 12V connection?



WEEK 6: SOLAR LOAD MANAGEMENT



Content: (I) Understanding electrical loads (II) Load prioritization and balancing (III) Estimating total load consumption (IV) Power rating and efficiency (V) Practical: Load distribution.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Electrical Load" and differentiate between AC and DC loads.
2. Differentiate between "Resistive," "Inductive," and "Phantom" loads.
3. Define "Load Prioritization" and explain its importance in an off-grid system.
4. Re-calculate the total load consumption (Wh/day) for a given household.
5. Create a simple load distribution plan for a home solar setup.

Lesson Content:

1. Understanding Electrical Loads

- **Definition:** An "electrical load" is *any* appliance or device that *consumes* electricity to do work (e.g., a bulb, a fan, a TV).
- **Load Estimation (Recap):** The process of calculating $\text{Power (W)} \times \text{Time (h)} = \text{Energy (Wh)}$.
- **Types of Loads:**
 - **AC Loads:** Appliances that use 230V AC (e.g., TVs, fans, fridges). They are powered by the *inverter*.
 - **DC Loads:** Appliances that use 12V/24V DC (e.g., DC bulbs, DC fans). They are powered *directly* from the battery/charge controller.
 - **Benefit:** Using DC loads is *more efficient* as it avoids the 10-15% energy loss from the inverter.

2. Detailed Load Types and Their Impact

- **A. Resistive Loads:**
 - **How they work:** Use electricity to create *heat* (e.g., a heating coil).
 - **Examples:** Incandescent light bulbs, electric irons, toasters, electric kettles.
 - **Impact:** They are "easy" loads for an inverter, *but* they consume *enormous* amounts of power (e.g., an iron is 1,200W+). They are the #1 enemy of a solar system and should be used sparingly, if at all.
- **B. Inductive Loads (Motor Loads):**
 - **How they work:** Use magnetic coils to create *motion*.
 - **Examples:** Fans, refrigerators, air conditioners, water pumps, blenders.
 - **Impact:** They have a high "**surge**" or "**startup**" current. A 150W fridge may need 600-800W for 1-2 seconds *just to turn on*.
 - **Rule:** Your inverter *must* be sized to handle this surge.
- **C. Phantom Loads (Vampire Loads):**

- **How they work:** Appliances that *secretly* consume small amounts of power *even when they are "OFF"* (on standby).
- **Examples:** A TV (waiting for the remote signal), a DSTV decoder, a microwave (powering the clock).
- **Impact:** These small loads add up over 24 hours and can drain your battery unnecessarily.
- **Solution:** Plug them into an extension strip and *switch them off completely* at the wall when not in use.

3. Load Prioritization and Balancing

- **Definition:** In an off-grid system, you have a *limited* amount of power (your battery). "Load prioritization" is the *conscious decision* of what is most important to power.
- **Pyramid of Priority:**
 - **Priority 1 (Critical Loads):** Must always be on. (e.g., Lights, phone chargers, security system).
 - **Priority 2 (Important Loads):** Used daily, but can be managed. (e.g., Fan, TV, laptop).
 - **Priority 3 (Luxury/Non-Essential Loads):** Use *only* when the sun is high and batteries are full. (e.g., Blender, sound system).
 - **Priority 4 (Forbidden Loads):** *Never* use on a small/medium solar system. (e.g., Electric iron, water heater, AC).
- **Load Balancing:** Not running all your heavy loads at the same time.
 - *Bad:* Turn on the TV, all lights, and the blender at the same time (inverter overload).
 - *Good:* Use the blender during the day (when sun is shining), and only the TV and lights at night.

4. Practical: Load Distribution Plan

- **Goal:** Create a "Load Distribution" or "Changeover" plan for a house with solar.
- **Process:**

1. Go to the main Distribution Board (DB) or "Fuse Box."
 2. Identify all the circuit breakers.
 3. Separate the house circuits into "Critical" and "Non-Critical."
 4. The **AC Output** of the **Inverter** is connected *only* to the "Critical" loads (e.g., lights, selected sockets).
 5. The "Non-Critical" loads (e.g., AC units, water heaters) are *left on the grid (NEPA) circuit*. They *cannot* be powered by the inverter.
- **Result:** When NEPA fails, the inverter *only* powers the essential items, preventing overload and making the battery last.

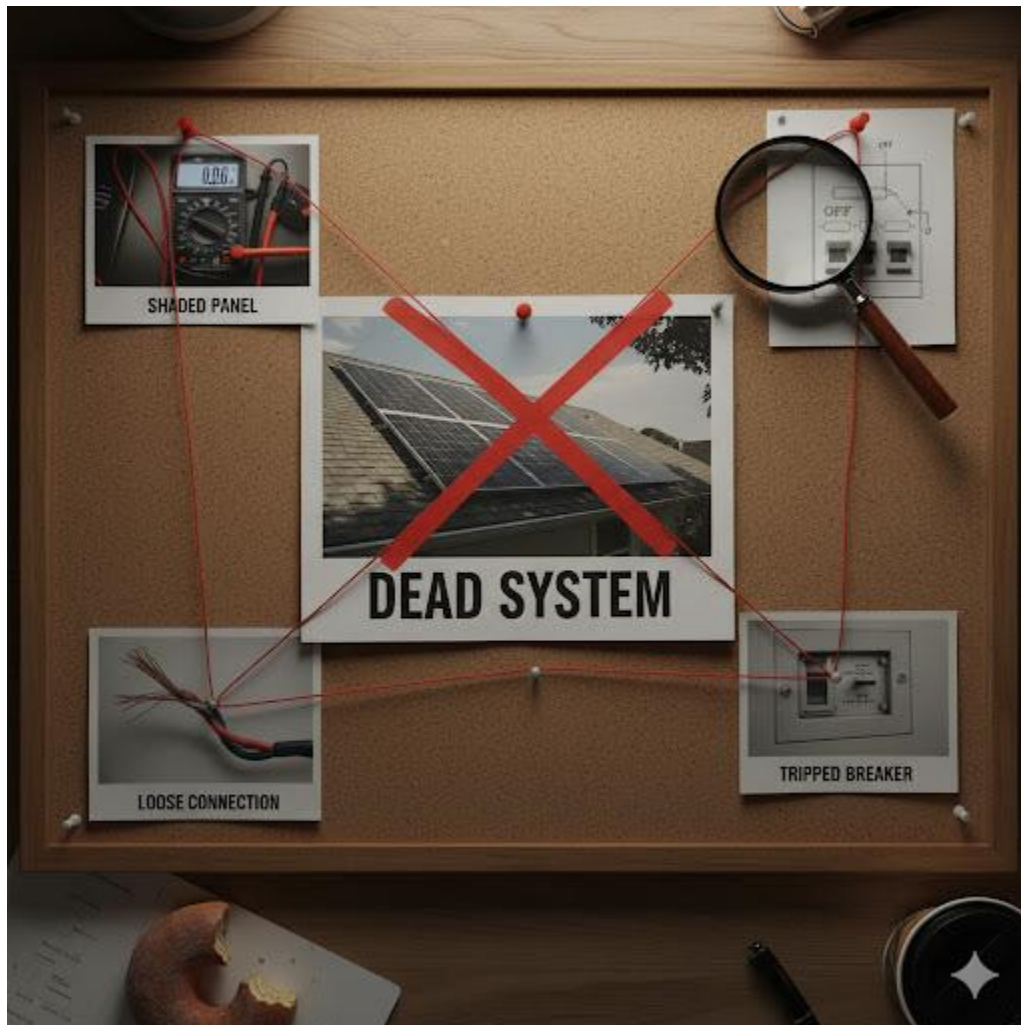
Evaluation:

1. What is an "electrical load"?
2. What is the main difference between a "resistive" load and an "inductive" load?
3. Why is an electric iron "forbidden" on most solar systems?
4. What is a "phantom load," and how do you stop it?
5. What is "load prioritization"?

Assignment:

1. Define: Resistive Load, Inductive Load, and Phantom Load.
2. Go through your house and list:
 - Two (2) resistive loads.
 - Two (2) inductive loads.
 - Two (2) phantom loads.
3. A customer has a 1.5kW inverter. They want to power a 150W fridge (with a 600W startup surge), a 100W TV, and a 1,200W electric iron.
 - Can they run all three at once? Why/why not?
 - What advice would you give them about "load balancing"?
4. What is a "load distribution board," and why is it important to separate "critical" and "non-critical" loads?
5. Why is it more efficient to use a 12V DC fan in a 12V solar system than a 230V AC fan?

WEEK 8: FAULT DETECTION AND TROUBLESHOOTING



Content: (I) Common faults in solar installations (II) Diagnostic tools (multimeter, tester, ammeter) (III) Identifying loose connections, faulty panels, and inverter issues (IV) Step-by-step troubleshooting (V) Practical: Testing a system.

Performance Objectives: By the end of this lesson, students should be able to:

1. List at least five (5) common faults in a PV system.
2. Demonstrate how to use a multimeter to measure DC Voltage (V) and Current (A).
3. Explain the "Open-Circuit Voltage (V_{oc})" and "Short-Circuit Current (I_{sc})" of a panel.
4. Develop a logical, step-by-step process for troubleshooting a "dead" system.
5. Demonstrate how to test for a faulty panel in a string.

Lesson Content:

1. Common Faults in Solar Installations

1. **Zero Output:** The system is completely dead.
2. **Low Output:** The system works, but batteries drain too fast or can't power what they used to.
3. **Inverter Beeping/Overload:** The inverter is protesting (see Week 6).
4. **Panel Issues:** Shading, dirt, or physical damage (cracks).
5. **Battery Failure:** The battery is old, sulphated, or has a dead cell.
6. **Wiring Issues: Loose connections** (at terminals), corrosion, or a broken wire.

2. Diagnostic Tools (The "Doctor's Kit")

- **Multimeter (The #1 Tool):** An essential device for measuring:
 - **DC Voltage (V):** To check panel and battery voltage.
 - **AC Voltage (V):** To check inverter output.
 - **Continuity (Beep Test):** To check if a wire is broken or a fuse is blown.
- **Clamp Ammeter (DC/AC):** A "clamp" that you put *around* a wire to measure the **Current (Amps)** flowing through it *without* disconnecting the wire. This is very useful for checking how many Amps the panels are producing.

3. Practical: How to Test a Solar Panel (Do this in the sun, with the panel *disconnected* from the system).

- **A. Open-Circuit Voltage (Voc) Test:**
 - **What it is:** The *maximum voltage* a panel can produce with *no load* connected.
 - **How:** Set your multimeter to **DC Voltage (V)**. Touch the Red probe to the panel's (+) connector and the Black probe to the (-) connector.
 - **Result:** A 12V panel should read between **18V - 22V** (this is the Voc). If it reads 0V or 5V, the panel is faulty.
- **B. Short-Circuit Current (Isc) Test:**
 - **What it is:** The *maximum current* a panel can produce.

- **How:** Set your multimeter to **DC Amps (A)** (e.g., 10A setting). Touch the probes to the (+) and (-) connectors.
- **Result:** This will show the Amps. If the panel is rated for 5A, it should read close to 5A in full sun. If it reads 0.5A, the panel is faulty.

4. Step-by-Step Troubleshooting Process

- **Customer Complaint: "My system is not working!"**
- **Rule:** Start from the *load* and work *backwards* to the *source*.
 - **Step 1: Check the Load.** Is the appliance (e.g., fan) turned on? Is it faulty? Try another appliance.
 - **Step 2: Check the Inverter.** Is the inverter ON? Are there any red lights or error codes (e.g., "Overload," "Low Batt")?
 - *If "Overload":* Tell them to unplug high-power loads (iron).
 - *If "Low Batt":* The problem is with *charging*.
 - **Step 3: Check the Battery (The Source).** Use a multimeter (DC-V) to test the battery bank voltage.
 - *Result 1: 10.5V. This is the problem.* The battery is *empty*. The system is not charging.
 - *Result 2: 12.5V.* The battery is *full*. The problem is the *inverter* (it's faulty).
 - **Step 4: (If Battery is Empty) Check the Charging.**
 - **A. Check Breakers:** Are the breakers between the panel and controller *ON*?
 - **B. Check Controller:** Is the charge controller's screen on? Does it show sunlight/charging?
 - **C. Check Panels:** Go to the charge controller's PV input terminals. Use a multimeter (DC-V) to test the voltage *coming from the panels*.
 - *Result 1: 20V.* Panels are good. The *controller* is faulty.
 - *Result 2: 0V.* The panels are *not* sending power. The problem is the *panels* or the *wiring* from the roof.
 - **Step 5: Check Panel Wiring.** Go to the roof. Check for loose MC4 connectors. Test each panel's Voc.

Evaluation:

1. List three common faults in a PV system.
2. What is the *most important* tool for troubleshooting?
3. What does "Voc" stand for, and how do you test it?
4. What is the *first* thing you should check if a customer says their system is "dead"?
5. A system is not charging. You test the battery, and it's 10.5V. You test the panel input at the controller, and it reads 20V. What component is most likely faulty? (The Charge Controller).

Assignment:

1. Define "Troubleshooting."
2. What is a "clamp ammeter" used for?
3. A 12V panel has a Voc of 21V and an Isc of 8A.
 - What setting do you put your multimeter on to test Voc?
 - What setting do you put it on to test Isc?
4. Draw a simple flowchart for troubleshooting a "low battery" problem (start from the battery and check backwards).
5. Why is it important to check for *loose connections* at the battery terminals? What can a loose connection cause?

WEEK 9: SOLAR SYSTEM TESTING AND COMMISSIONING



Content: (I) Procedures for testing system performance (II) Checking output voltage and current (III) Performance indicators (efficiency, shading) (IV) Final commissioning checklist (V) Group project: Commissioning a setup.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Commissioning" as the final step of an installation.
2. State the purpose of a commissioning test.
3. Perform a full system test (checking panel Voc, battery voltage, and inverter output V-AC).
4. Explain performance indicators like "shading effects" and "battery health."

5. Create and follow a final commissioning checklist.

Lesson Content:

1. What is System Commissioning?

- **Definition:** "Commissioning" is the formal process of *testing* and *verifying* every part of a newly installed solar system to ensure it is working **correctly, safely, and efficiently** *before* you "hand over" the system to the customer.
- **Purpose:**
 1. To guarantee **safety** (no loose wires, proper grounding).
 2. To guarantee **performance** (the system is producing the power you promised).
 3. To create **documentation** (a record) that the job was completed correctly.

2. Procedures for Testing System Performance This is a step-by-step verification using your multimeter.

- **Step 1: Visual Inspection (The "Look" Test):**
 - Are all panel clamps tight?
 - Is all wiring neat and secured in trunking (conduits)?
 - Are all breakers and components clearly *labeled* (e.g., "PV IN," "BATTERY")?
 - Is the battery room ventilated?
 - Are all safety warnings posted?
- **Step 2: Polarity and Voltage Checks (The "Measure" Test - *with loads OFF*):**
 - **Panel Array:** Disconnect from controller. Test Voc (e.g., 48V). Check polarity.
 - **Battery Bank:** Disconnect from system. Test voltage (e.g., 25V). Check polarity.
- **Step 3: System "Boot-up" (The "Connection" Test):**
 - Connect in the correct order: (1) Battery to Controller, (2) Panel to Controller, (3) Battery to Inverter.
 - Check all controller and inverter screens. Are there any error codes?
- **Step 4: Performance Check (The "Load" Test):**
 - **Inverter Output:** Turn on the inverter. Use a multimeter (AC-V) to check the output socket. It should read ~220-240V AC.

- **Load Test:** Turn on the customer's loads (e.g., fan, TV) one by one. Does the inverter handle them?
- **Charging Test:** Use a DC clamp ammeter to measure the *current (Amps)*:
 - Flowing from the panels to the controller.
 - Flowing from the controller to the battery. (This confirms the system is charging).

3. Key Performance Indicators

- **Efficiency:** Is the system producing the power it was *designed* to? (This is related to the Amps you just measured).
- **Shading Effects:** A major performance *killer*. You must do a final check. Is there a new TV antenna, tree branch, or neighboring building that casts a shadow on the panels between 9 am and 3 pm?
- **Battery Health:** Is the battery voltage stable? Does it drop quickly under load?

4. Final Commissioning Checklist A checklist is a professional tool. You tick off every item, then you and the customer sign it.

Commissioning Checklist	Done (✓)
A. Safety	
1. All metal frames (panels, inverter) are grounded.	[]
2. All circuit breakers and fuses are installed correctly.	[]
3. All wires are in trunking (no exposed wires).	[]
4. All connections are tight (no loose terminals).	[]
5. Battery room is ventilated.	[]
B. Performance	
1. Panel array Voc is correct (___ V).	[]
2. Battery bank voltage is correct (___ V).	[]
3. Inverter AC output is correct (___ V).	[]
4. System charges battery (Solar light on controller is ON).	[]
5. System powers all "Critical Loads" correctly.	[]

C. Handover	
1. All components are labeled.	[]
2. Customer has been trained on how to use the system.	[]
3. Customer has the maintenance schedule and emergency contact.	[]
Installer Signature: _____	Customer Signature: _____

5. Group Project: Commissioning a Classroom Setup

- (Students are given a pre-installed (or pre-faulted) solar setup).
- **Task:** In groups, go through the *entire* commissioning checklist.
- **Goal:** Use multimeters and visual checks to fill out the checklist and declare the system "Safe and Operational" or "Faulty" (and state the fault).

Evaluation:

1. What is "commissioning" a solar system?
2. What is the purpose of a commissioning checklist?
3. What are the three (3) main *categories* on a commissioning checklist?
4. What tool would you use to check the AC output voltage of an inverter?
5. What is the very *last* step of a commissioning process? (Handover/Training the customer).

Assignment:

1. Define "Commissioning."
2. Why is the "Visual Inspection" step so important for safety?
3. Why do you need to *label* all the breakers for the customer?
4. What is a "shading effect," and why is it a "performance killer"?
5. Create your own simple 5-point commissioning checklist for the small DC lighting system you designed in Week 3.

WEEK 10: DOCUMENTATION AND REPORTING



Content: (I) Recording installation data (II) Report writing and job documentation (III) Preparing invoices and customer service records (IV) Group task: Prepare a mock project report.

Performance Objectives: By the end of this lesson, students should be able to:

1. Explain the importance of documentation in business.
2. Differentiate between a "Quotation" and an "Invoice."
3. List the key information that must be included in a final "Job Report."
4. Create a simple, professional invoice for a solar installation.
5. Prepare a mock project report for a given installation.

Lesson Content:

1. Importance of Documentation

- **Definition:** Documentation is the process of *keeping written records* of all your business activities.
- **Importance:**
 1. **Professionalism:** It shows the customer you are an organized and serious business.
 2. **Legal Protection:** It is *proof* of what you agreed to do, what you did, and what the customer paid. This protects you in any dispute.
 3. **Future Maintenance:** It provides a *history* of the system, making it *much* easier to troubleshoot faults 5 years later.
 4. **Financials:** You need it to track your profit, loss, and pay your taxes.

2. Key Business Documents

- **A. The Quotation (Before the Job):**
 - (Recap from JSS3 Term 1) This is your *offer* to the customer. It details the *price* and *work* to be done.
- **B. The Invoice (After the Job):**
 - **Definition:** This is the *bill*. It is a formal, written *request for payment* for a job you have *completed*.
 - **Content:** It looks similar to a quotation but is a *demand*, not an *offer*. It must include:
 - Your business name and contact info.
 - The customer's name.
 - **Invoice Number** and **Date**.
 - A list of services/goods *provided*.
 - **Total Amount Due**.
 - **Payment Terms** (e.g., "Due on receipt" or "Payment due in 30 days").
- **C. The Receipt (After Payment):**

- **Definition:** This is *proof of payment*. You give this to the customer *after* they have paid the invoice.
- **D. The Job Report (Handover Document):**
 - **Definition:** A technical document for the customer that *details* the system you installed.
 - **Content:**
 - Customer name and installation date.
 - **System Components:** A full list (e.g., 4 x 320W *Monocrystalline* Panels, 1 x 3.5kW *PSW* Inverter, 2 x 12V 200Ah *Gel* Batteries).
 - **System Diagram:** A simple wiring diagram.
 - **Commissioning Checklist:** The *signed* checklist from Week 9.
 - **Warranty Information:** Details on the warranty for each component.
 - **Maintenance Schedule:** The checklist you created in Week 8.

3. Group Task: Prepare a Mock Solar Project Report

- **Task:** Based on the "Classroom Setup" project from Week 9, groups will write a 1-page "Job Report."
- **Sections:**
 1. **Project:** Classroom Lighting System
 2. **Date:** [Today's Date]
 3. **Components Installed:** (e.g., 1 x 50W Panel, 1 x 10A Controller, 1 x 30Ah Battery)
 4. **Commissioning Results:** (e.g., Panel Voc: 21V, Battery: 12.8V, Inverter: N/A, Load: 3 Bulbs working)
 5. **Maintenance Advice:** (e.g., "Clean panel once a month").

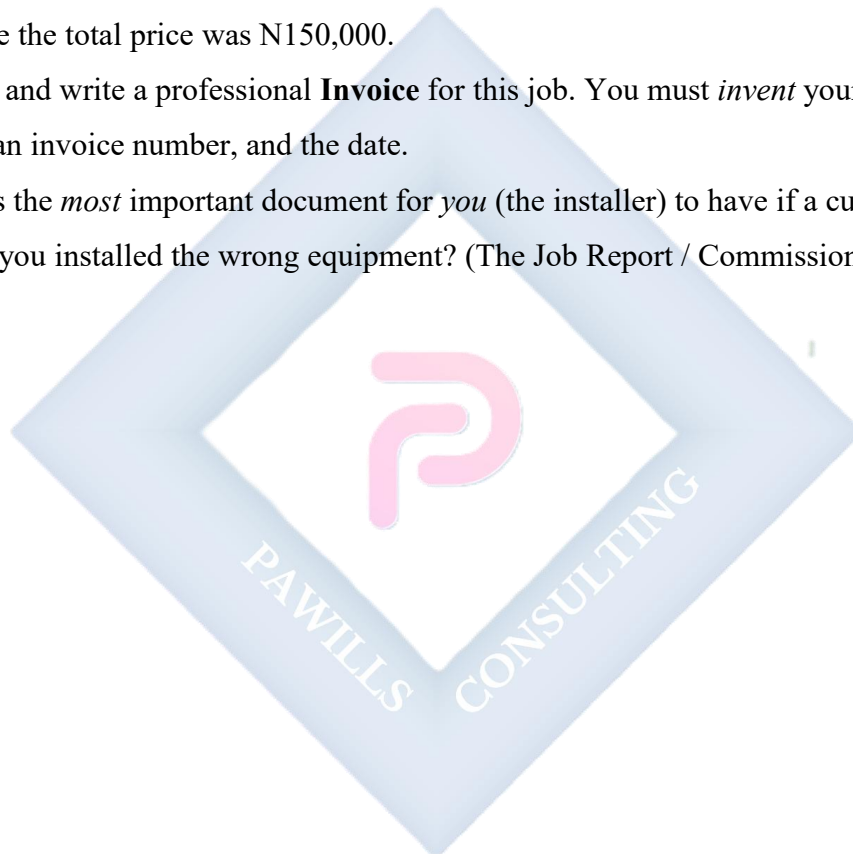
Evaluation:

1. Why is documentation important for "legal protection"?
2. What is the main difference between a "Quotation" and an "Invoice"?
3. What is a "Receipt"?

4. List three (3) key pieces of information that must be in a "Job Report."
5. Why is it important to give the customer a maintenance schedule?

Assignment:

1. Define: Quotation, Invoice, Receipt, and Job Report.
2. You completed the system for the student in the Week 3 assignment (Laptop + 2 Bulbs).
3. Assume the total price was N150,000.
4. Design and write a professional **Invoice** for this job. You must *invent* your company name, an invoice number, and the date.
5. What is the *most* important document for *you* (the installer) to have if a customer falsely claims you installed the wrong equipment? (The Job Report / Commissioning Checklist).



Week 11: Revision

Topic: Revision

Content: Comprehensive review of Second Term topics.

Note: This week is dedicated to a comprehensive review of all topics covered in the Second Term, especially in preparation for the end-of-term examination.

Revision Checklist:

1. First Term Review (Week 1):

- Do you know the 3 configurations (Standalone, Grid-tied, Hybrid)?
- Do you know the difference between MPPT/PWM and Li-ion/Lead-Acid?
- Can you do a 4-step system sizing (Load, Inverter, Battery, Panel)?

2. Electrical Wiring (Week 2):

- AC vs. DC cables (where are they used?).
- What is "Polarity," and what are the DC colours (Red/Black)?
- What is "Cable Sizing," and why is it a fire-safety issue?
- What is "Voltage Drop"?
- What is the correct *order* of connecting components? (Controller->Batt, then Panel->Controller).

3. Battery Bank (Week 3):

- **Series (+ to -):** Voltage ADDS, Ah is same.
- **Parallel (+ to +, - to -):** Ah ADDS, Voltage is same.
- **Compound:** How to build a 24V, 400Ah bank (4 batteries).
- Factors that *kill* batteries (Heat, High DoD, No maintenance).

4. Inverter (Week 4):

- Function (DC to AC).
- PSW (Safe, clean) vs. MSW (Dangerous, dirty).
- Safety: **NO Reverse Polarity**, use thick/short cables, use a fuse, ventilate.

5. Load Management (Week 6):

- Resistive (heat - iron), Inductive (motor - fridge), Phantom (standby - TV).

- Why are Inductive loads hard for an inverter? (Surge current).
- "Load Prioritization" (Critical vs. Non-Critical).

6. Troubleshooting (Week 8):

- Main tool: Multimeter.
- How to test a panel (Voc, Isc).
- The logical process: Load -> Inverter -> Battery -> Controller -> Panels.

7. Commissioning (Week 9):

- Definition: Final safety and performance check.
- The Checklist: Safety, Performance, Handover.
- Testing: Voc, V-batt, V-AC, Amps charging.

8. Documentation (Week 10):

- Quotation (Offer) vs. Invoice (Bill) vs. Receipt (Proof of Payment).
- Job Report (Handover document).

Activity:

- Comprehensive Q&A session.
- Final practical exam: Given a set of components, students must (1) assemble it, (2) test it, and (3) fill out a commissioning checklist.

**THIRD TERM - JSS 3 SOLAR PHOTOVOLTAIC (PV) INSTALLATION AND
MAINTENANCE LESSON NOTES**

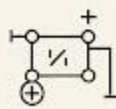


WEEK 1: REVISION OF PREVIOUS WORK

The Solar Installer's Toolbelt: Knowledge & Skills



Sizing



Series/Parallel



Troubleshooting



Safety

Content: (I) Review of solar components, wiring, and maintenance (II) Discussion on real-life solar installation scenarios (III) Introduction to professional ethics in solar practice.

Performance Objectives: By the end of this lesson, students should be able to:

1. Recall and summarize the entire solar installation process from sizing to commissioning.
2. Confidently identify all major solar components and their functions.
3. Discuss common real-world installation challenges and propose solutions.
4. Define "Professional Ethics" and list three ethical principles for an installer.

Lesson Content:

1. Comprehensive Review of JSS 3 (Terms 1 & 2) This week serves as a capstone review, connecting all practical skills learned.

- **A. System Sizing & Components (Recap):**

- **Sizing:** The 4-step process: (1) Load Calculation (W & Wh), (2) Inverter Sizing (W), (3) Battery Sizing (Ah), (4) Panel Sizing (W).
- **Components:**
 - **Panels:** Mono vs. Poly
 - **Controllers:** PWM vs. MPPT (MPPT is ~30% more efficient).
 - **Batteries:** Lead-Acid (50% DoD) vs. Li-ion (80-100% DoD).
 - **Inverters:** MSW (Modified - dangerous) vs. PSW (Pure - safe).

- **B. Wiring & Connections (Recap):**

- **Safety:** Polarity (Red +, Black -). Reversing it destroys equipment.
- **Sizing:** Cables must be thick enough for the current (Amps) to prevent *fire*.
- **Drop:** Voltage drop is the power lost over *long* cables.
- **Series (+ to -):** Voltage Adds, Ah is Same.
- **Parallel (+ to +, - to -):** Ah Adds, Voltage is Same.

- **C. Maintenance & Troubleshooting (Recap):**

- **Maintenance:** Preventive (cleaning panels) vs. Corrective (fixing faults).
- **Troubleshooting:** The logical process: Load -> Inverter -> Battery -> Controller -> Panels.
- **Tools:** Multimeter (for Voc, V-batt) and Clamp Ammeter (for Isc, A-charge).

2. Discussion: Real-Life Installation Scenarios (Teacher presents scenarios, students propose solutions)

- **Scenario 1:** "A customer has a 12V system. Their batteries are 50 meters away from the inverter. The inverter keeps beeping 'Low Battery' even when the batteries are full."
 - **Problem:** Extreme **Voltage Drop**. The 12V is "lost" over the 50m cable.
 - **Solution:** (1) Move the batteries *much* closer (< 1.5m). (2) Use *extremely* thick (and expensive) cables. (3) The *best* solution: re-design the system to 24V or 48V, which is less affected by voltage drop.

- **Scenario 2:** "A customer's system (installed 6 months ago) works for only 1 hour at night, down from 5 hours."
 - **Problem:** Battery failure.
 - **Possible Causes:** (1) **Undersized Panels:** The panels are too small to ever fully charge the batteries. (2) **Sulphation:** The batteries were left empty for too long. (3) **Overload:** The customer added a new freezer without telling you. (4) **High Heat:** The batteries are stored in a hot, unventilated room.
- **Scenario 3:** "A customer calls you one day after installation. The charge controller is smoking, and the inverter is dead."
 - **Problem:** Catastrophic failure.
 - **Most Likely Cause:** The installer **reversed the polarity** (+ to -, - to +) on the controller or inverter. This is an irreversible, costly mistake.

3. Introduction to Professional Ethics in Solar Practice

- **Definition:** "Ethics" are the moral rules and principles that guide your behavior as a professional. In solar, your reputation is built on **Trust**.
- **Key Ethical Principles for an Installer:**
 1. **Honesty & Transparency:**
 - **NEVER** lie about products. Don't sell a 200W panel as a 300W panel.
 - **NEVER** sell a "Modified" sine wave inverter to a customer who needs to power a TV or fridge.
 - Be clear about what the system *can* and *cannot* do (e.g., "This system *cannot* power your electric iron").
 2. **Safety First (Above All):**
 - Your primary duty is to ensure the system is safe for the customer.
 - **NEVER** cut corners on safety (e.g., "Grounding takes too long, let's skip it." -> This is highly unethical).
 - **NEVER** use thin, cheap wires that can cause a fire.
 3. **Competence & Quality:**
 - Do a *neat* and *professional* job. Manage your cables, label your breakers.

- Do not take a job you are not qualified for (e.g., a 100kW factory-scale installation).
- **Own your mistakes.** If something goes wrong, don't blame the customer. Fix it.

4. Confidentiality:

- You are working inside a customer's home. Respect their privacy. Do not discuss their property or family with others.

Evaluation:

1. What are the four steps of system sizing?
2. What is the main danger of "Reversed Polarity"?
3. A customer's new system (with Lead-Acid batteries) fails after 1 year. You discover the panels are too small and the batteries are always at 20% DoD. What is the most likely cause of failure? (Sulphation due to chronic undercharging).
4. What does "Professional Ethics" mean?
5. What is the *most* unethical thing an installer can do regarding safety? (e.g., Not grounding the system, using undersized wires).

Assignment:

1. In your own words, what is the difference between a "real-life" scenario and a "textbook" scenario? (Hint: Textbooks are perfect, real-life has problems like rust, rats, and unexpected shading).
2. What is the *most* important ethical principle for a solar installer, and why?
3. A customer with a 1kW system (that *cannot* power an iron) asks you, "Can I use my iron just for 5 minutes?" What is the *honest* and *ethical* answer?
4. Draw a compound connection for four 12V, 200Ah batteries to make a 24V, 400Ah bank.
5. What is the primary danger of *not* using a charge controller and connecting a panel directly to a battery?

WEEK 2: OCCUPATIONAL HEALTH AND SAFETY (OHS) IN SOLAR WORK



Content: (I) Importance of safety (II) Common electrical hazards and prevention (III) Safe handling of panels, batteries, and tools (IV) Personal Protective Equipment (PPE) (V) Practical: Safety drill and PPE inspection.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Occupational Health and Safety" (OHS).
2. Identify the three (3) main categories of hazards in solar work (Electrical, Falls, Chemical).
3. List the essential Personal Protective Equipment (PPE) for a solar installer.
4. Explain the procedure for "Lockout/Tagout" (LOTO).

5. Demonstrate safe handling of batteries and tools.

Lesson Content:

1. Importance of OHS

- **Definition: Occupational Health and Safety (OHS)** is a set of rules and practices designed to protect the health and safety of *workers* (you, the installer) and *the public* (the customer) from workplace-related injuries and accidents.
- **Motto:** "Safety First." No job is so urgent that you can't take the time to do it safely. It is always better to go home *safe* than to finish a job *fast*.

2. Common Hazards in Solar Installation

- **A. Electrical Hazards (The Most Dangerous):**
 - **Source 1: DC Shock:** Solar panels in series can produce 100-600V DC. This high-voltage DC is *more dangerous* than AC. It can stop your heart.
 - **Source 2: AC Shock:** The output of the inverter (230V AC) can cause severe shock.
 - **Source 3: Arc Flash:** A high-amperage short circuit (like dropping a metal tool on a battery bank) creates an *explosion* of molten metal and light. It can cause fatal burns and blindness.
- **B. Fall Hazards (The Most Common):**
 - **Source:** Working on slanted, slippery, or weak roofs.
 - **Risk:** A fall from a one-storey building can lead to permanent disability or death.
 - **Prevention:** Using ladders correctly (3-points of contact), using a safety harness, and never working on a roof in the rain.
- **C. Chemical Hazards:**
 - **Source:** Lead-acid batteries.
 - **Risk:** The electrolyte is **sulfuric acid**, which causes severe chemical burns to skin and *permanent blindness* if it splashes in the eyes.
 - **Risk 2:** Charging batteries release explosive **hydrogen gas**.
- **D. Manual Handling Hazards:**

- **Source:** Carrying heavy panels, inverters, and batteries.
- **Risk:** Back injury from improper lifting.

3. Personal Protective Equipment (PPE) PPE is your "last line of defense."

- **1. For All Jobs:**
 - **Safety Boots:** Steel-toed, rubber-soled. Protects from falling objects and provides grip.
 - **Hand Gloves:** Rubber-coated grip gloves for handling panels.
- **2. For Electrical Work:**
 - **Insulated Gloves:** Special gloves rated for 1000V.
 - **Insulated Tools:** Screwdrivers and spanners with 1000V-rated handles.
 - **Safety Glasses:** To protect from sparks.
- **3. For Battery Work:**
 - **Safety Goggles (Chemical):** *Essential*. Must seal around the eyes.
 - **Acid-Resistant Gloves:** (e.g., Nitrile or rubber gloves).
- **4. For Roof Work:**
 - **Safety Harness:** A harness and rope system that "catches" you if you fall.

4. Hazard Prevention Procedures

- **A. Lockout / Tagout (LOTO):**
 - **Definition:** A formal procedure to ensure that dangerous equipment (like a circuit breaker) is **properly shut off and cannot be turned on again** while you are working on it.
 - **How it works:**
 1. **Lockout:** You turn the breaker OFF and put *your own padlock* through the handle.
 2. **Tagout:** You attach a *tag* with your name and "DANGER: DO NOT OPERATE."
 - **Result:** No one else can accidentally turn the power on while your hands are on the wires.

- **B. Safe Handling:**

- **Panels:** Do not carry on your head. Carry vertically with a partner. Do not walk on panels.
- **Batteries:** Lift with your legs, not your back. Keep upright.
- **Tools:** Never use tools with broken or cracked insulation.

5. Practical: Safety Drill and PPE Inspection

- **Activity 1:** Teacher shows various PPE (hard hat, safety glasses, goggles, insulated gloves, grip gloves). Students must identify what each one is for.
- **Activity 2:** Teacher demonstrates the "**3-points of contact**" rule for climbing a ladder (e.g., two feet and one hand *always* on the ladder).
- **Activity 3:** Teacher demonstrates the LOTO procedure on a circuit breaker.

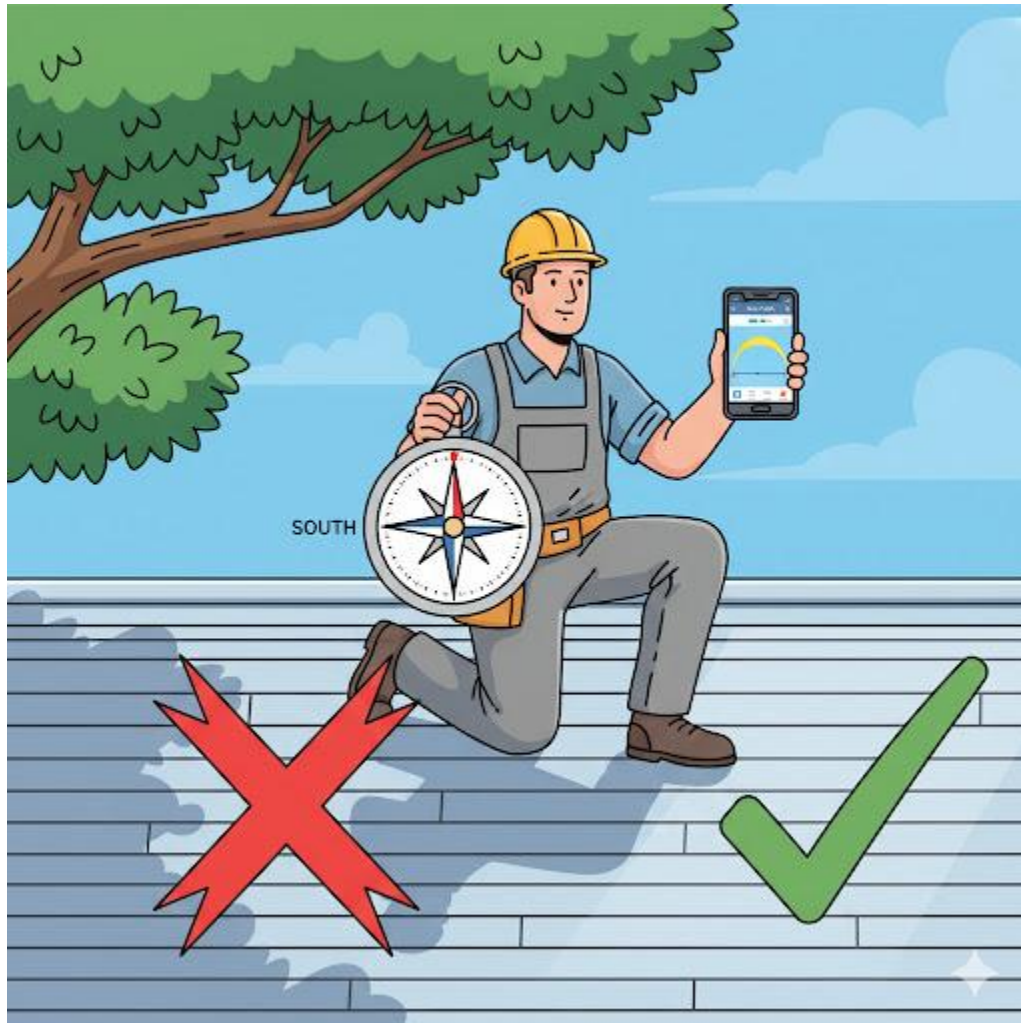
Evaluation:

1. What does "OHS" stand for?
2. What are the three (3) main categories of hazards in solar installation?
3. What is the *most* important piece of PPE to wear when working with lead-acid batteries? (Chemical Goggles).
4. What is an "Arc Flash"?
5. What is the purpose of "Lockout/Tagout" (LOTO)?

Assignment:

1. What is the "Golden Rule" of OHS? ("Safety First").
2. Why is high-voltage DC (from panels) considered more dangerous than AC? (Hint: It's harder to let go).
3. What are the "3 points of contact" when climbing a ladder?
4. Why should you *never* smoke or create sparks near a charging lead-acid battery?
5. Draw a simple "Tag" for a "Tagout." What three things should be written on it? (e.g., DANGER, your name, date).

WEEK 3: SOLAR SITE ASSESSMENT



Content: (I) Identifying suitable locations (II) Factors affecting site selection (orientation, shading, tilt angle, roof strength) (III) Site survey checklist and measurements (IV) Practical: Performing a basic site assessment.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Solar Site Assessment" and state its purpose.
2. Identify the optimal *orientation* (direction) and *tilt angle* for panels in Nigeria.
3. Define "Shading Analysis" and explain why it is the most critical part of an assessment.
4. List the key items on a site survey checklist.
5. Perform a basic site assessment for a given building (e.g., the school).

Lesson Content:

1. What is a Solar Site Assessment (Survey)?

- **Definition:** A site assessment is a formal visit to the customer's location *before* an installation to gather all the data needed to design a safe, efficient, and cost-effective system.
- **Purpose:** This is the *first step* of any professional installation. You cannot design a system or write a quotation (Term 1) without a proper site survey.
- **Goal:** To answer the question: "Is this site good for solar, and where is the *best* place to put the equipment?"

2. Factors Affecting Site Selection (Panel Location)

- **A. Orientation (Azimuth) - The Most Important Factor:**
 - **Definition:** The compass direction the panels will face (N, S, E, W).
 - **Rule:** The sun rises in the East, travels across the sky, and sets in the West.
 - **In Nigeria (Northern Hemisphere):** To get the maximum all-day sun exposure, panels must *always* face **TRUE SOUTH**.
 - **Compromise:** East-facing roofs are "okay" (they get strong morning sun). West-facing roofs are also "okay" (they get strong afternoon sun).
 - **Avoid: Never** install on a North-facing roof in Nigeria. It will be in shadow most of the day.
- **B. Shading Analysis (The Second Most Important Factor):**
 - **Definition:** Identifying *anything* that will cast a shadow on the panels, *at any time of day, at any time of year*.
 - **Sources of Shade:** Trees, neighboring buildings, satellite dishes, water tanks, chimneys, other parts of the roof.
 - **Impact: Shade is a performance killer.** If even *one cell* on a panel is shaded, the power output of the *entire panel* can drop by 50-100%. If panels are in series, shading one panel can shut down the *entire string*.

- **Rule:** Panels *must* be installed in a location that is **100% shadow-free** between 9:00 a.m. and 3:00 p.m.
- **C. Tilt Angle:**
 - **Definition:** The "steepness" of the panel relative to the flat ground.
 - **Rule:** The "perfect" tilt angle is usually equal to your latitude. For Nigeria (latitude $\sim 4^\circ$ to 14° N), a shallow angle of **10 to 15 degrees** is optimal.
 - **Practical:** Most roofs already have a tilt. If it's a flat roof, you *must* use "tilt-mounts" to angle the panels up.
- **D. Roof Strength and Type:**
 - **Inspection:** Is the roof made of corrugated iron, aluminum, or concrete? Is it strong enough to hold the weight of the panels and installers? Are there any leaks or broken rafters?

3. Site Survey Checklist (What to record) (An installer arrives at the site with a form, a compass, and a tape measure)

- **1. Customer Needs:**
 - What are your "critical loads"? (Week 2)
 - What is your budget?
 - Where do you want the batteries/inverter to be?
- **2. Roof Assessment:**
 - Roof type (metal, concrete) and condition?
 - **Orientation** (What direction does it face? - *Use a compass app on a phone*).
 - **Shading** (Any trees/buildings? - *Take photos*).
 - **Measurements** (How many meters long/wide is the available space? - *Use a tape measure*).
- **3. Equipment Location:**
 - Where will the inverter and batteries go? (Must be a *cool, dry, ventilated* space, e.g., a garage or store, *not* a hot kitchen).
- **4. Cable Run:**
 - What is the distance from the roof (panels) to the inverter (controller)? (This is for calculating *voltage drop* - Week 2).

4. Practical: Performing a Basic Site Assessment

- **Activity:** Take the students outside to assess a school building.
- **Task 1 (Orientation):** Use a compass (or phone app) to determine which direction the main roof faces (e.g., "This roof faces South-East").
- **Task 2 (Shading):** Stand back and look at the roof. (e.g., "There is a large mango tree that will shade the roof after 2 p.m. This is a bad location").
- **Task 3 (Equipment):** Identify a good, safe, cool location for the batteries and inverter (e.g., "The principal's store room").
- **Task 4 (Measurement):** Estimate the cable run from the roof to the store room (e.g., "About 20 meters").

Evaluation:

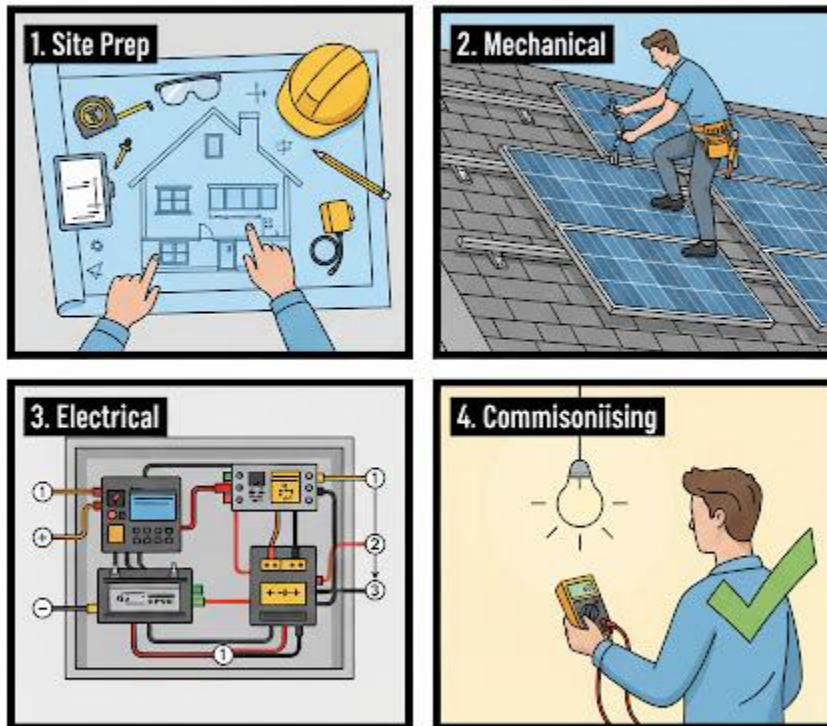
1. What is a "Solar Site Assessment"?
2. What is the *most important* factor in a site assessment? (Shading).
3. In Nigeria, what is the *optimal* direction (orientation) for solar panels to face?
4. Why is a "shallow" tilt angle (10-15 degrees) used in Nigeria?
5. Why is it a very bad idea to install batteries in a hot kitchen? (Heat kills batteries).

Assignment:

1. Define Orientation, Shading, and Tilt Angle.
2. Why is shading on even *one part* of a panel so bad for the whole system?
3. You arrive at a customer's house. The *only* available roof faces North. What do you tell the customer?
4. Create a simple 5-point "Site Survey Checklist" for your own use.
5. Perform a *basic* site assessment of your own house or apartment building (if safe to do so).
 - (a) What direction does the best roof face?
 - (b) Are there any shading issues?
 - (c) Where would be a good, cool place to put the batteries and inverter?

WEEK 4: SYSTEM INSTALLATION PROCEDURE

SYSTEM INSTALLATION PROCEDURE



Content: (I) Step-by-step installation process (mounting, wiring, connections, testing) (II) Mounting structures (roof and ground) (III) Installing all components (IV) Practical: Installing a small off-grid system.

Performance Objectives: By the end of this lesson, students should be able to:

1. List the major steps of a solar installation in the correct chronological order.
2. Differentiate between a "roof mount" and a "ground mount."
3. Describe the process of wiring the main components (panels, controller, battery, inverter).
4. Participate in the practical installation of a small solar lighting system.

Lesson Content:

This lesson combines all the skills we have learned into a single, step-by-step procedure.

1. Step-by-Step Installation Process (Overview)

1. Phase 1: Preparation (Site & Safety)

- (Week 3) Conduct Site Assessment.
- (Term 1) Finalize System Design and Quotation.
- Gather all tools, components, and (Week 2) PPE.

2. Phase 2: Mechanical Installation (The "Build")

- (Week 4, Term 1) Install mounting structures (roof or ground).
- Mount the solar panels onto the rails.
- Install the "Balance of System" (BOS): inverter, charge controller, battery bank in their designated safe, cool location.

3. Phase 3: Electrical Installation (The "Wiring")

- (Week 2) Run PV cables from panels to controller (in conduit).
- (Week 3) Assemble the battery bank (series/parallel).
- (Week 2) Make all final connections in the *correct order*.
- (Week 4, Term 1 / Week 2, Term 3) Install all safety devices (breakers, grounding).

4. Phase 4: Commissioning & Handover (The "Test & Finish")

- (Week 9, Term 2) Test all voltages, currents, and loads.
- (Week 10, Term 2) Train the customer and hand over the documentation.

2. Mounting Structures

• A. Roof Mounting (Most Common):

- **Method:** Uses rails and clamps bolted to the roof rafters.
- **Pros:** Uses "dead space," is secure, and is cost-effective.
- **Cons:** Risk of roof leaks if not sealed properly, difficult to access for cleaning, limited by roof size/orientation.

• B. Ground Mounting:

- **Method:** A strong metal frame (using poles set in concrete) is built on the ground to hold the panels.
- **Pros:**
 1. **Optimal Orientation:** You can face it *perfectly* South, regardless of the roof.
 2. **Optimal Tilt:** You can set the *perfect* 10-15 degree angle.
 3. **Easy Maintenance:** Very easy to access for cleaning.
- **Cons:** Uses valuable land, more expensive (requires concrete and steel), at risk of theft or vandalism if not secured.

3. Installing Components (The Flow of Work)

- **On the Roof (Mechanical):**

- Measure and mark rafter locations.
- Install mounting feet with sealant.
- Attach rails to feet.
- Run grounding wire to all rails.
- Mount panels one by one, securing with mid/end clamps.
- Ground each panel frame to the rail.
- Connect the panels (series/parallel) using MC4 connectors.
- Run the final PV "home-run" cable (in conduit) down to the inverter room.

- **In the "Inverter Room" (Electrical):**

1. Mount the inverter, charge controller, and breakers securely to the wall (in a ventilated, accessible spot).
2. Place the battery bank in its *final position* (on a raised stand, *not* on the bare floor).
3. Run the grounding wire from the inverter chassis to the main earth rod.
4. **Make the connections (as per Week 2):**
 - **1st:** Battery Bank -> Breaker -> Charge Controller (BATT terminals)
 - **2nd:** Battery Bank -> Main DC Breaker -> Inverter (DC Input)
 - **3rd:** PV Array Cable -> PV Breaker -> Charge Controller (PV terminals)

- **4th:** Inverter (AC Output) -> AC Breaker -> AC Loads (Distribution Board)

4. Practical: Installing a Small Off-Grid Solar Lighting System

- **Goal:** Assemble the system designed in Week 3, Term 1 (12V, 36W+ system).
- **Components:** 1 x 50W panel, 1 x 10A PWM controller, 1 x 12V 30Ah battery, 2 x 12V DC bulbs, wires, breakers, and tools.
- **Group Task:**
 1. **Group 1 (Mounting):** Securely mount the 50W panel on a frame (outside, facing South).
 2. **Group 2 (BOS):** Mount the charge controller and 12V breaker on a "practice board."
 3. **Group 3 (Wiring):**
 - Run the PV cable from the panel to the controller (do *not* connect yet).
 - Connect the battery to the 12V breaker.
 - Connect the breaker to the controller (BATT terminals). (Controller lights up).
 - Connect the DC bulbs to the "LOAD" terminals on the controller.
 - **Final Step:** Connect the PV cables to the "PV" terminals. (Charging light comes on).
 4. **Testing:** Use the controller's button (if available) to turn on the DC bulbs. Use a multimeter to test V-panel, V-batt.

Evaluation:

1. What are the four (4) main phases of a solar installation?
2. What is the main *advantage* of a ground-mounted system over a roof-mounted one?
3. What is the main *disadvantage* of a ground-mounted system?
4. When wiring the "inverter room," what is the *first* component you must connect to the charge controller?

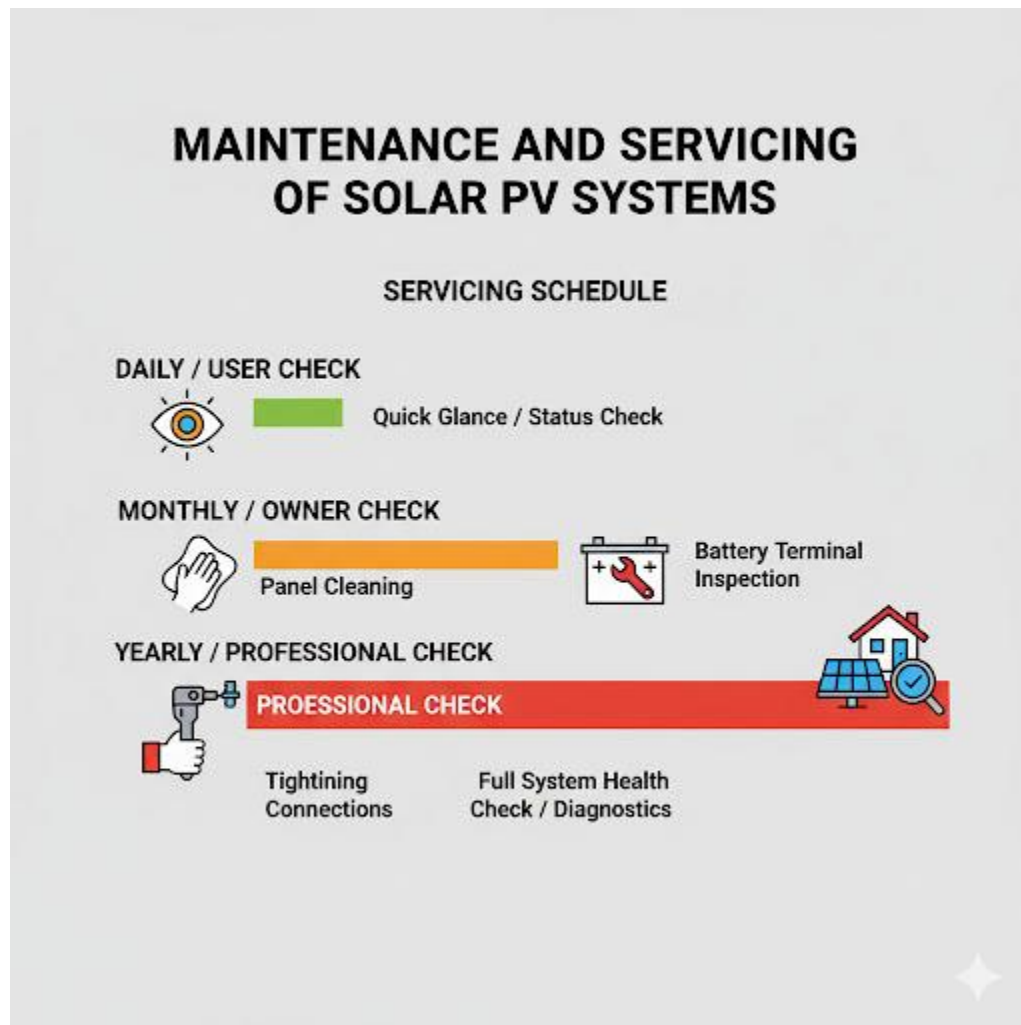
5. Why should batteries *not* be placed directly on a bare concrete floor? (Hint: It drains their temperature and charge).

Assignment:

1. List the 4 phases of installation (e.g., Preparation, Mechanical...).
2. Draw a simple diagram of a roof-mounted panel, showing the **Roof, Rafter, Mounting Foot, Rail, and Panel**.
3. What are two "pros" and two "cons" of a roof-mounted system?
4. Why must you use a *sealant* when drilling holes in a roof?
5. You have a 48V inverter system. You have four 12V panels. What connection (series or parallel) *must* you use for the panels to match the system? (Series).



WEEK 6: MAINTENANCE AND SERVICING OF SOLAR PV SYSTEMS



Content: (I) Types of maintenance (preventive, corrective) (II) Cleaning panels and maintaining batteries (III) Checking connections (IV) Maintenance schedules (V) Practical: Conducting maintenance.

Performance Objectives: By the end of this lesson, students should be able to:

1. Differentiate between Preventive and Corrective Maintenance.
2. Explain the correct, safe procedure for cleaning solar panels.
3. Describe the key maintenance checks for a lead-acid battery bank.
4. Create a simple annual maintenance schedule.
5. Perform a basic maintenance check on a live system (inspect, clean, test).

Lesson Content:

1. Types of Maintenance (Recap)

- **A. Preventive Maintenance (Proactive):**
 - **Definition:** Routine, scheduled checks and servicing you perform *before* a problem happens.
 - **Goal:** To prevent faults, maximize efficiency, and extend the life of the system.
 - **Example:** Cleaning dirty solar panels.
- **B. Corrective Maintenance (Reactive):**
 - **Definition:** Repair work you perform *after* a component has failed.
 - **Goal:** To fix a problem and get the system back online.
 - **Example:** The inverter is dead; you must replace it.
- **Key Idea:** Good Preventive Maintenance (which costs very little) saves a huge amount of money on Corrective Maintenance.

2. Maintenance Procedure: Solar Panels

- **Why:** Dust, pollen, bird droppings, and soot can block sunlight and **reduce your panel's power output by up to 30%.**
- **When:**
 - Visually inspect monthly.
 - Clean every 6-12 months (or more in very dusty areas like the Harmattan).
 - **Safety:** Clean *only* in the **early morning or late evening.**
 - **DANGER:** Never spray cold water on a *hot* panel (in the 1 p.m. sun). The "thermal shock" can crack the glass.
- **How:**
 1. Turn off the system (trip the PV breaker) for safety.
 2. Use a soft brush (with a long extension pole) and clean water.
 3. A *very* mild, non-abrasive soap (like car wash soap) can be used.
 4. *Never* use harsh chemical detergents (like Omo) or abrasive (scratchy) scouring pads.
 5. Rinse thoroughly and allow to dry.

3. Maintenance Procedure: Battery Bank

- **Safety:** Always wear PPE (goggles and gloves).
- **A. For All Batteries (Monthly):**
 - **Cleanliness:** Wipe the tops of the batteries. Dust and moisture can create a small path for electricity to "leak," slowly discharging them.
 - **Connections:** Check that all terminals are **tight**. A loose terminal *will* overheat, melt, and cause a fire or system failure.
 - **Corrosion:** Look for a white, powdery substance on the terminals. Clean it off with a battery cleaning solution (or a mixture of baking soda and water), then re-tighten and apply terminal grease.
- **B. For Flooded Lead-Acid Batteries (Monthly/Quarterly):**
 - **Check Water Level:** Carefully open the vent caps.
 - **Inspect:** The water (electrolyte) level *must* be *above* the lead plates.
 - **Top-Up:** If low, add **distilled water** (battery water) only. **NEVER add tap water** (it will kill the battery) and **NEVER add more acid**.

4. Maintenance Schedule (Annual Service by Installer)

- **1. Visual:** Check all components for damage, rust, or leaks.
- **2. Mechanical:** Check that all panel mounts and clamps are still tight.
- **3. Cleaning:** Clean solar panels and battery terminals.
- **4. Electrical (The Most Important):**
 - **Torque Check:** *Re-tighten* all major electrical connections (at breakers, controller, inverter) with an insulated screwdriver. They can become loose over time due to vibration and temperature changes.
 - **Voltage Check:** Test Voc of panels and resting voltage of batteries to ensure they are healthy.

5. Practical: Conducting Maintenance

- **Task:** Groups are assigned to the school's solar system (or a lab setup).
- **Checklist:**

1. (Safety) Put on PPE. Perform LOTO (Week 2).
2. (Panel) Visually inspect panels for dirt/cracks. (Simulate cleaning).
3. (Battery) Inspect terminals. Are they clean? Are they tight? (Show teacher).
4. (Electrical) Open the controller/inverter. Check that all wire connections are secure.
5. (Log) Record all findings in a "Maintenance Logbook."

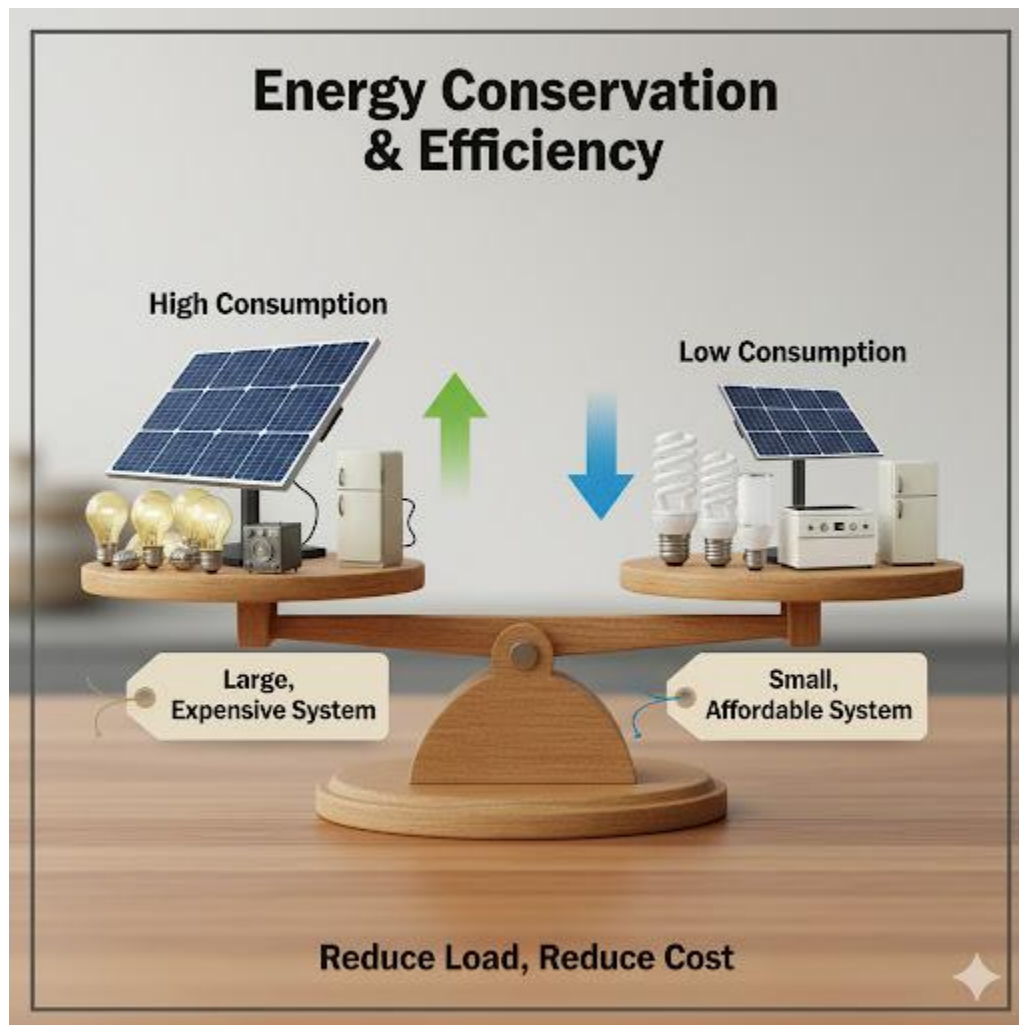
Evaluation:

1. What is the difference between Preventive and Corrective Maintenance?
2. Why should you *never* clean a solar panel at 1 p.m.?
3. What is the main danger of a *loose* battery terminal?
4. What liquid *must* you use to top-up a flooded lead-acid battery?
5. What is a "Torque Check"?

Assignment:

1. Define Preventive and Corrective Maintenance.
2. What PPE is required for (a) cleaning panels on a roof and (b) servicing batteries?
3. What is "corrosion" on a battery terminal, and how do you clean it?
4. Create a simple "Annual Maintenance Checklist" for an installer (at least 5 items).
5. Why should you *never* use tap water in a battery? (Research this).

WEEK 8: ENERGY CONSERVATION AND EFFICIENCY



Content: (I) Meaning of energy conservation (II) Ways to reduce power consumption (III) Importance of energy-efficient appliances (IV) Role of solar in sustainable living (V) Group discussion.

Performance Objectives: By the end of this lesson, students should be able to:

1. Differentiate between "Energy Conservation" and "Energy Efficiency."
2. List at least five (5) ways to conserve (reduce) energy in the home.
3. Explain the importance of "Energy Star" or energy-efficient appliances.
4. Explain how using solar technology contributes to sustainable living.
5. Design a simple "energy-efficient home" plan.

Lesson Content:

1. Definitions

- **A. Energy Conservation:**

- **Definition:** This is a change in *behavior* that results in using *less* energy.
- **Simple Meaning:** "Using less."
- **Example:**
 - **Turning off** a light when you leave the room.
 - **Unplugging** a "phantom load" (like a TV) from the wall.
 - **Reducing** the number of hours you watch TV.

- **B. Energy Efficiency:**

- **Definition:** This is using *technology* that requires *less* energy to perform the *same* job.
- **Simple Meaning:** "Wasting less."
- **Example:**
 - **Replacing** an old 100W incandescent (yellow) bulb with a new 10W **LED** bulb. (You get the *same* amount of light, but use 90% less energy).
 - **Replacing** an old, power-hungry CRT TV with a modern, flat-screen **LED** TV.

2. Ways to Reduce Power Consumption (Conservation)

- This is the *cheapest* and *easiest* way to make your solar system's battery last longer.
1. **Turn off lights** when not in the room.
 2. **Turn off fans** when not in the room.
 3. **Unplug "Phantom Loads":** Unplug TVs, decoders, and phone chargers from the wall when not in use.
 4. **Manage Loads (Load Balancing):** Don't run the fridge, TV, and all fans at the same time.
 5. **Use Natural Light:** Open curtains during the day instead of turning on a light.

3. Importance of Energy-Efficient Appliances

- **The "Secret" to Solar:** A solar system is expensive. The *smartest* way to save money is to *first* reduce your load, so you can buy a *smaller, cheaper system*.
- **The "Enemy" Loads:** Resistive loads (irons, kettles, heaters) and old, inefficient appliances (old fridges, incandescent bulbs).
- **The "Hero" Loads:**
 - **LED Bulbs:** Use ~90% less energy than old bulbs.
 - **Inverter Appliances:** Modern "Inverter Fridges" or "Inverter ACs" use *smart* motors that consume 30-50% less energy than old ones.
 - **"Energy Star" Rating:** Look for appliances with an "Energy Star" label. This is a certification that proves the appliance is highly efficient.
- **Impact on Sizing:**
 - *Bad Design:* A house with 10 x 60W old bulbs = 600W load.
 - *Good Design:* A house with 10 x 7W LED bulbs = 70W load.
 - The "Good Design" house can use a *much smaller* (and cheaper) inverter and battery bank to get the *same* amount of light.

4. Role of Solar in Sustainable Living

- **Sustainable Living:** (Recap from JSS3 Social Studies) - Living in a way that does not deplete resources for future generations.
- **How Solar Helps:**
 1. **Environmental:** It produces zero CO₂, zero air pollution, and zero noise. It directly fights *climate change* (SDG 13).
 2. **Economic:** It provides energy independence and reduces (or eliminates) electricity bills.
 3. **Social:** It provides reliable power for schools, hospitals, and homes, improving education, health, and quality of life.

5. Group Discussion: Designing an Energy-Efficient Home

- **Task:** If you were building a *new* house designed to be "solar-friendly," what 5 things would you do?
- **Answers:**
 - Use only LED bulbs.
 - Buy *only* inverter fridges/appliances.
 - Install DC (12V) fans instead of AC fans.
 - *Forbid* electric irons/kettles (use a gas iron/kettle).
 - Install large windows for natural light.
 - Insulate the roof to keep the house cool (reduces need for fans/AC).

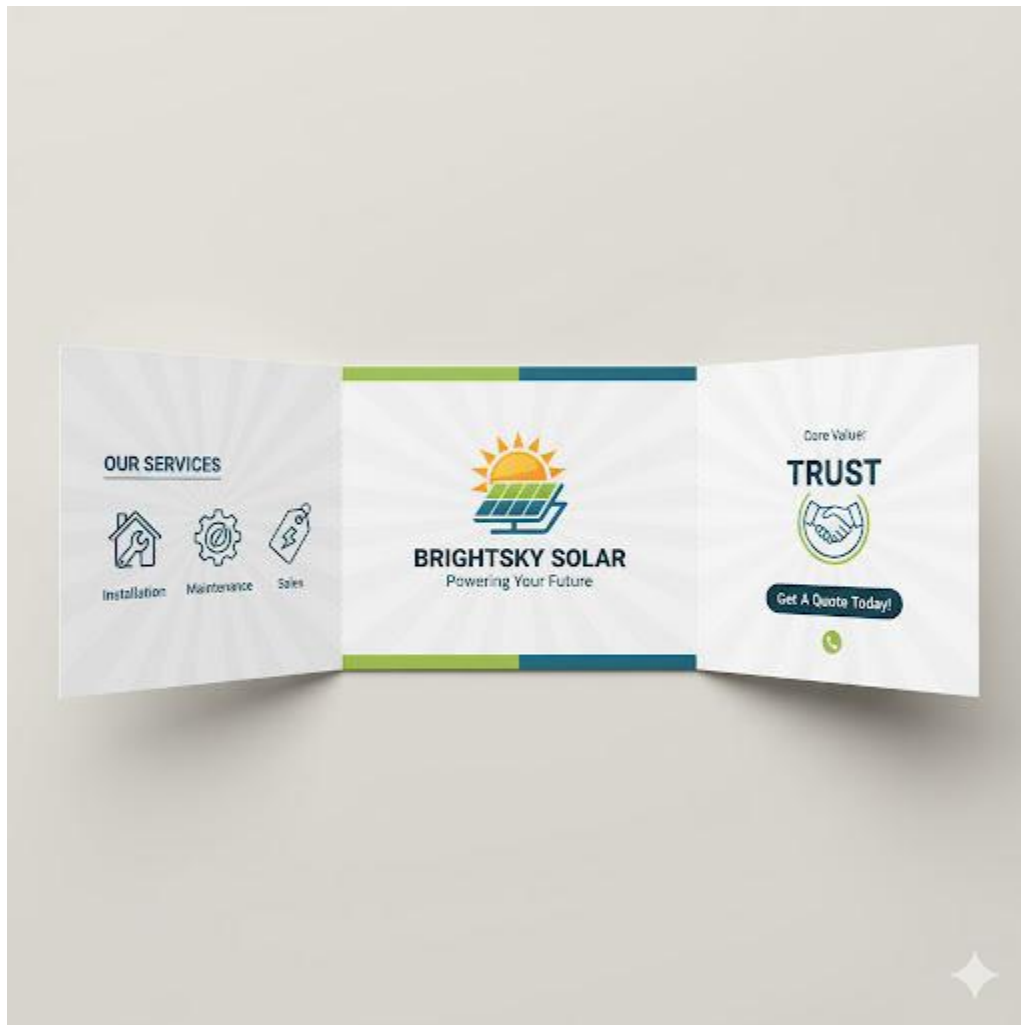
Evaluation:

1. What is the difference between "Energy Conservation" and "Energy Efficiency"?
2. Is "turning off a light" an example of conservation or efficiency?
3. Is "replacing an old bulb with an LED" an example of conservation or efficiency?
4. What is a "phantom load"?
5. Why is it "smarter" to buy an "Inverter Fridge" for a solar system?

Assignment:

1. Define Energy Conservation and Energy Efficiency.
2. Do a "load audit" of your living room. List all appliances and label them as "Resistive," "Inductive," or "Phantom."
3. What are "Energy Star" appliances?
4. List three (3) ways *you* can conserve energy at home this week.
5. How does using solar power help achieve "Sustainable Living"?

WEEK 9: ENTREPRENEURSHIP IN SOLAR ENERGY



Content: (I) Understanding solar business opportunities (II) Types of solar services (installation, maintenance, consultancy, sales) (III_ Basic costing and quotation writing (IV) Building customer relations (V) Project: Develop a simple solar business plan.

Performance Objectives: By the end of this lesson, students should be able to:

1. Recall the definition of "Entrepreneurship" (JSS2).
2. Identify four (4) distinct business opportunities (career paths) in the solar industry.
3. Differentiate between "Cost," "Price," and "Profit."
4. Write a simple, professional quotation for a solar installation.
5. Develop a one-page "Solar Business Plan."

Lesson Content:

1. Entrepreneurship in Solar (Recap)

- **Entrepreneur:** A person who *identifies a need* (e.g., "People need reliable power"), *organizes a business* to meet that need, and *takes a financial risk* to do so.
- **The Opportunity:** Nigeria has a massive *power deficit* and abundant *sunshine*. This creates one of the biggest business opportunities in the country.

2. Types of Solar Services (Career Paths)

1. **Solar Installer:** (The most common)
 - **Job:** Designs and physically *installs* complete systems for homes and businesses.
 - **Skills:** Technical (sizing, wiring, OHS), problem-solving.
2. **Solar Retailer / Wholesaler (Sales):**
 - **Job:** Buys components (panels, batteries) in *bulk* from manufacturers and *sells* them to installers or the public.
 - **Skills:** Marketing, sales, inventory management, finance.
3. **Solar Maintenance Technician:**
 - **Job:** Offers "Maintenance Contracts." They *do not* do new installations; they *service* existing ones (cleaning, testing, repairs) for a yearly fee.
 - **Skills:** Troubleshooting, customer service, reliability.
4. **Solar Consultant / Designer:**
 - **Job:** An *expert* who does *not* install. They are hired to do the "brain" work: perform the site assessment, do the advanced sizing, and create the engineering diagrams. The installer then follows their plan.
 - **Skills:** Advanced engineering, design software, communication.

3. Basic Costing and Quotation Writing

- **A. Costing (Your Cost):**
 - **Cost:** The amount of money *you* (the entrepreneur) spend to do the job.

- Total Cost = Cost of Components (panels, batteries, etc.) + Cost of Logistics (transport) + Cost of Labor (paying your workers)
- **B. Pricing (The Customer's Price):**
 - **Price:** The amount the customer pays *you*.
 - **Profit:** The money you make.
 - **Formula:** Price = Total Cost + Profit (Markup)
 - *Example:* Your Total Cost is N500,000. You add a 20% profit (N100,000). The Price you put on the quotation is **N600,000**.
- **C. Quotation (Review):**
 - A formal, written *offer* to the customer. It must be professional and *itemized* (show the price for each component) to build trust. (See Term 1, Week 10 for example).

4. Building Customer Relations and Trust

- In an industry with many "quacks," **TRUST** is your #1 marketing tool.
- 1. **Be Honest:** Do not lie about component capacity. Do not sell MSW as PSW.
- 2. **Be Professional:** Show up on time, wear a clean t-shirt with your logo, and do a *neat* installation (cable management).
- 3. **Be Safe:** Use proper PPE and grounding. A customer who sees you are *safe* will *trust* you.
- 4. **Educate the Customer:** After installation, *train* them on how to use their system, what the inverter lights mean, and how to manage their loads (Week 6). This prevents 90% of future "faulty" calls.

5. Project: Develop a Simple Solar Business Plan

- **A Business Plan:** A document that outlines your business goals and how you will achieve them.
- **Group Task:** Create a 1-page "Solar Business Plan."
 - **1. Business Name:** (e.g., "Sunshine Solar Solutions")
 - **2. Slogan:** (e.g., "Reliable Power, Honest Service")
 - **3. Business Type:** (Choose one: Installer, Retailer, or Maintenance)

- **4. Target Market:** (Who are your customers? e.g., "Small businesses and new homeowners in my LGA")
- **5. My "Brand" (How I build trust):** (e.g., "We will build trust by offering a 1-year free maintenance check and by *showing* the customer the specs of every component before we install it.")

Evaluation:

1. What is an "entrepreneur"?
2. List three (3) different business opportunities in the solar industry.
3. What is the difference between "Cost," "Price," and "Profit"?
4. What is the *most* important thing an installer is selling (even more than panels)? (Trust).
5. What is a "Quotation"?

Assignment:

1. Define Entrepreneurship.
2. Choose one of the four solar career paths. Write one paragraph explaining why you would (or would not) like that job.
3. What is the difference between a "Solar Retailer" and a "Solar Installer"?
4. Write a simple, 1-page "Solar Business Plan" for your *own* (imaginary) company, following the 5-point structure in the lesson.
5. Why is it important to *educate* the customer after an installation?

WEEK 10: PROJECT WORK AND FIELD DEMONSTRATION



Content: (I) Group project: Design, install, and test a mini solar setup (II) Documentation: materials, cost, performance (III) Oral presentation (IV) Evaluation and peer review.

Performance Objectives: By the end of this lesson, students should be able to:

1. Work effectively in a group to solve a practical problem.
2. Apply all skills learned (Sizing, Safety, Installation, Wiring, Commissioning) to a real project.
3. Create a complete "Project Report" documenting their work.
4. Present their project, process, and results to the class.
5. Critically evaluate the work of other groups.

Lesson Content:

1. Project Brief: The "JSS 3 Capstone Project"

- **Goal:** To design, install, test, and document a small, functional, standalone solar PV system from start to finish.
- **Project:** "The Solar-Powered Classroom Hub"
- **Load:**
 - 2 x 12V DC LED Bulbs (7W each)
 - 1 x 12V DC Phone Charging Station (with 4 USB ports, ~10W)
- **Task:** Each group must build a complete, safe, and functional system to power this load.

2. Phase 1: Design and Sizing (Group Work)

- (Students must do this *before* building)
- **1. Load Calculation:**
 - Bulbs: $(2 \times 7W) \times 4 \text{ hours/day} = 14W \times 4h = 56 \text{ Wh/day}$
 - Charging: $10W \times 6 \text{ hours/day} = 60 \text{ Wh/day}$
 - **Total Load:** $56 + 60 = 116 \text{ Wh/day}$
- **2. Inverter Sizing: None.** (This is a 100% DC system).
- **3. Battery Sizing (12V):**
 - $(116 \text{ Wh} \times 1 \text{ day}) / (0.50 \text{ DoD}) = 232 \text{ Wh}$
 - $Ah = 232 \text{ Wh} / 12V = 19.3 \text{ Ah}$
 - **Selection:** 1 x 12V, 20Ah or 30Ah Battery.
- **4. Panel Sizing:**
 - $116 \text{ Wh} / (5 \text{ PSH} \times 0.70) = 116 / 3.5 = 33.1 \text{ W}$
 - **Selection:** 1 x 40W or 50W Solar Panel.
- **5. Controller Sizing:**
 - $Amps = 50W / 12V = 4.16 \text{ A}$
 - **Selection:** 1 x 5A or 10A PWM Controller.

3. Phase 2: Installation and Commissioning (Practical)

- **Materials (per group):** 1 panel, 1 controller, 1 battery, 2 DC bulbs, 1 USB port, wires, breakers, tools, multimeter.
- **Process:**
 1. **Safety First:** Put on PPE.
 2. **Mount:** Mount the controller, breakers, and bulb holders on a "project board."
 3. **Wire (Order):**
 - (1) Battery -> Breaker -> Controller (BATT)
 - (2) Panel -> Breaker -> Controller (PV)
 - (3) DC Loads (Bulbs/USB) -> Breaker -> Controller (LOAD)
 4. **Test (Commission):**
 - Use a multimeter to test Panel Voc and Battery V.
 - Turn on breakers.
 - Check controller screen.
 - Test the "Load" button to turn on the lights.
 - Plug in a phone to test the USB port.

4. Phase 3: Documentation and Presentation (Group Work)

- **A. Documentation (The Job Report):** Each group must submit a 1-page report containing:
 1. **Group Members:**
 2. **Project Goal:** (e.g., "To power lights and charging").
 3. **Sizing Calculations:** (Show your math for Wh, Ah, and Panel W).
 4. **Final Components List:** (e.g., "1x 40W Panel, 1x 10A PWM...")
 5. **Wiring Diagram:** (A simple, neat drawing).
 6. **Commissioning Results:** (V-panel, V-batt, "Did it work? Y/N").
- **B. Oral Presentation:**
 - Each group has 5 minutes to:
 1. Present their project board.
 2. Explain their design choices.
 3. Demonstrate that it works (turn on the light).

4. Discuss any challenges they faced (e.g., "We reversed the polarity by accident and had to fix it").

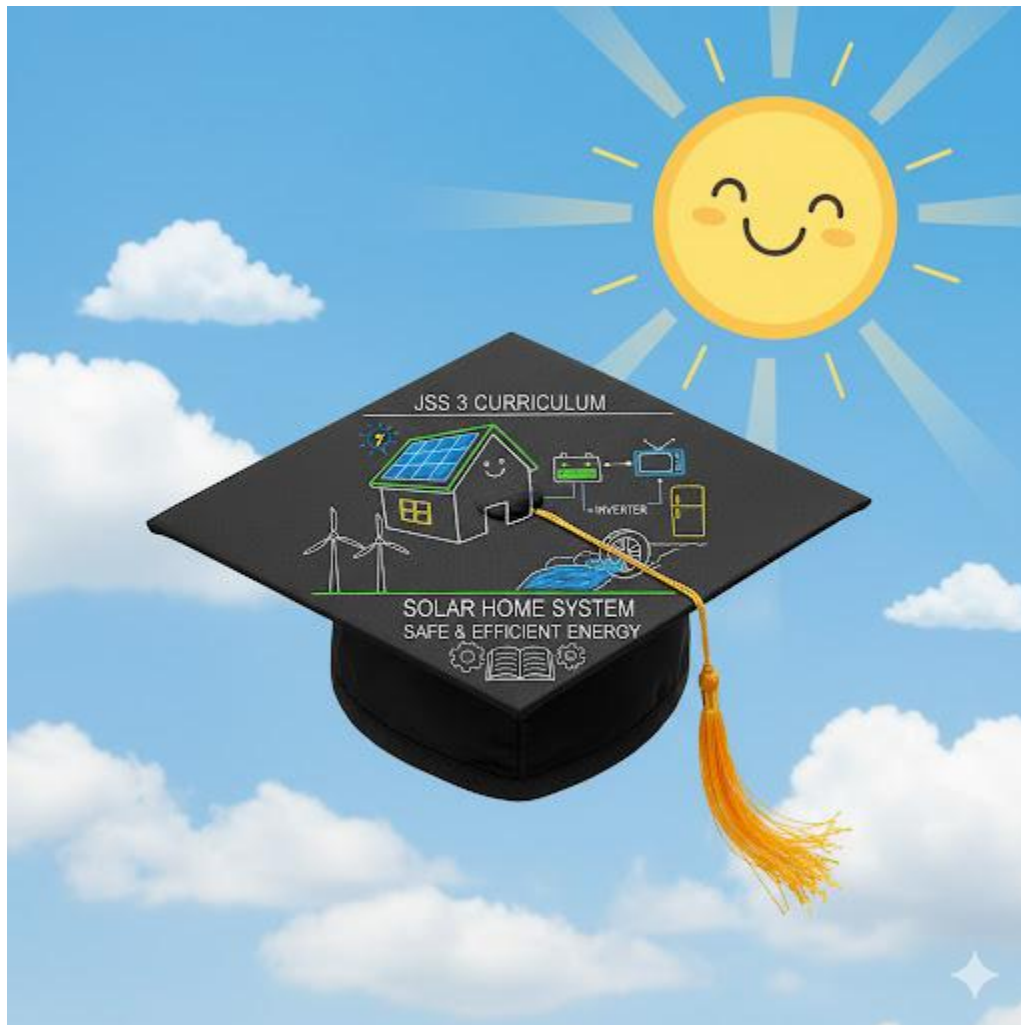
Evaluation: This project is the main assessment for the term.

- **Design (25%):** Were the sizing calculations correct?
- **Installation (25%):** Was the wiring neat, safe, and in the correct order?
- **Functionality (25%):** Did the system work?
- **Documentation & Presentation (25%):** Was the report clear and the presentation professional?

Assignment:

1. Complete your group's "Job Report" for the project.
2. Write one paragraph on what was the *biggest challenge* your group faced and how you solved it.
3. Write one paragraph on what you would do *differently* if you did the project again.
4. What is one "real-world" application for the mini-system you just built? (e.g., for a security post, a small shop, a rural reading light).
5. What is the most important thing you learned from this project?

WEEK 11: REVISION



Content: General revision of the year's topics and practical exercises.

Note: This week is dedicated to a comprehensive review of the *entire* JSS 3 curriculum, preparing students for their final BECE practical and theory examinations.

Revision Checklist (Entire JSS 3):

- **Term 1:**
 - **Advanced Components:** Can you *explain* (not just list) Mono/Poly, Li-ion/Lead-Acid, MPPT/PWM, PSW/MSW?

- **Advanced Sizing:** Can you *calculate* the W, Wh, Ah, and Panel size for a given load?
- **Connections:** Can you *draw* Series, Parallel, and Compound connections?
- **Safety:** Can you *list* the safety devices and their functions (Breaker, Fuse, Grounding)?
- **Maintenance:** Can you *create* a maintenance checklist?
- **Business:** Can you *write* a simple quotation?
- **Term 2:**
 - **Wiring:** Do you know the *order of connection*? Do you know the *danger of polarity*? Do you know *why* we use thick cables (high current, low voltage drop)?
 - **Battery Banks:** Can you *calculate* the final V and Ah for any connection? Do you know what *kills* batteries (Heat, High DoD)?
 - **Inverters:** Do you know the *safety rules* (no reverse polarity, ventilation, fuse)?
 - **Load Management:** Can you identify *Resistive, Inductive, Phantom* loads? Do you know what "load prioritization" is?
 - **Troubleshooting:** Can you use a *multimeter*? Do you know what *Voc* and *Isc* are? Can you *logically* find a fault (Load -> Invert -> Batt -> Cont -> Panel)?
 - **Commissioning:** What is a *commissioning checklist* and what is it for?
 - **Documentation:** What is an *Invoice* vs. a *Job Report*?
- **Term 3 (This Term):**
 - **OHS:** What are the 3 main hazards (Falls, Shock, Chemical)? What is *LOTO*?
 - **Site Assessment:** What are the key factors? (*Shading, Orientation, Tilt*).
 - **Installation:** What is the difference between *roof* and *ground* mounting?
 - **Energy Conservation:** What is *Conservation* (behavior) vs. *Efficiency* (technology)?
 - **Entrepreneurship:** What are the 4 solar *career paths*? How do you build *trust*?

Activity:

- **Final Practical Exam:** Students (in new groups) are given a "faulty" system (e.g., a wire is disconnected). They have 30 minutes to:
 1. Troubleshoot the system.

2. Identify the fault (e.g., "The panel breaker was off," "The battery polarity was reversed").
 3. Fix the fault and commission the system.
 4. Fill out a commissioning checklist.
- **Theory:** Review BECE past questions.

